

Poincaré Duality Groups

Anthony Dender

July 13, 2026

Contents

1	Introduction	2
2	Preliminaries	3
2.1	Poincaré Duality for Closed Manifolds	3
2.2	Relative Poincaré for Compact Manifolds	4
2.3	Closed Surfaces	4
3	Group Homology	5
3.1	Definition	5
3.2	Topological Interpretation	5
3.3	Dimension and Finiteness Properties	6
4	Duality Groups	7
4.1	Poincaré Duality Groups	7
4.2	Duality Groups	7
5	Group Pairs	9
5.1	Relative Homology and Cap Products	9
6	Trees and Bass–Serre Theory	12
6.1	Trees and Partial Orders	12
6.2	Splittings and Bass–Serre Theory	14
7	Almost Invariant Sets and Stallings’ Theorem	14
7.1	Almost Invariant Sets and Ends	14
7.2	Weight	16
7.3	Stallings’ Theorem and Cut Sets	17
8	Simultaneous Splitting	19
8.1	Proof of Theorem 8.1	20
8.2	Construction of Trees	24
9	Proof of the Main Theorem	26
10	Applications / Further Questions	34

1 Introduction

A closed oriented n -manifold M satisfies Poincaré duality: taking the cap product with a fundamental class $[M] \in H_n(M; \mathbb{Z})$ gives isomorphisms

$$H^k(M; A) \cong H_{n-k}(M; A)$$

for suitable coefficients A . If M is not orientable, the same statement holds in a suitable local coefficient system. Poincaré duality is one of the central homological features of closed manifolds.

Group homology allows one to ask for an analogous property in a purely algebraic setting. If M is aspherical and $G = \pi_1(M)$, then M is a $K(G, 1)$ complex, so the homology and cohomology of M agree with the group homology and cohomology of G . In this way, the Poincaré duality isomorphisms of M become isomorphisms in the homology of the group G . This motivates the definition of a *Poincaré duality group*: a PD^n group is a group whose homology behaves like that of the fundamental group of a closed aspherical n -manifold.

In [20] (problem G2, p. 391), C.T.C. Wall first famously asked whether every Poincaré duality group arises as the fundamental group of a closed aspherical manifold (a $K(G, 1)$ manifold). It was later proven that this statement holds for Poincaré duality groups of dimension two, giving the main theorem of this essay:

Theorem 1.1. *A group is a PD^2 group if and only if it is a surface group.*

Here, a *surface group* is a group isomorphic to the fundamental group of an aspherical closed surface. This remarkable result was first proven decisively in 1983, in the paper [9] by Beno Eckmann and Peter Linnell, which built upon a great deal of work on the topic by Eckmann and his students Robert Bieri and Heinz Müller.

The most technical input in the proof is a simultaneous splitting theorem of Müller [13], which is in the spirit of Stallings' theorem on ends [15]. Roughly, Müller's theorem constructs compatible splittings of a group G and a subgroup $T \leq G$ from suitable almost invariant subsets. The theorem depends on the *weight* of T relative to a group pair $(G; S_1, \dots, S_m)$, which measures how complicated the interaction is between T , G , and the associated subgroups S_i . The applications needed for the proof of Theorem 1.1 only involve weights 1 and 2, and so this essay proves only the corresponding restricted form of Müller's theorem.

This essay will be split into three main parts. First will be a brief overview of Poincaré duality in the case of compact manifolds with and without boundary (Section 2), followed by an overview of the analogous definitions in group homology and an account of their relevant properties (Sections 3-5). Second will be an overview of the background information and useful theories necessary to the proof, including the main theorem of Bass–Serre theory (Section 6) and the theory of almost invariant sets (Section 7). Finally, a proof of the main theorem of [13] in our restricted setting (Section 8), followed by a proof of the main theorem (Section 9), will be given.

The proof of the theorem given in Section 9, in the order in which it is presented here, will proceed roughly as follows. Beginning with a PD^2 group G :

1. Prove that G has a cyclic factor group, by showing that $\beta_1(G) > 0$. (Theorem 9.1)
2. Establish that G splits over a finitely generated subgroup L , and that L must be free of finite rank. (Corollary 9.1)

3. Prove that, given a splitting of G over a free group L of rank > 1 , there exists a splitting of G over a free group of strictly lower rank than L . (Lemma 9.3)
4. Conclude by induction that G splits over an infinite cyclic group C . Using the fact that C is a PD^1 group, obtain PD^2 pairs $(G_1; C)$ and $(G_2; C)$ or $(G; C, p^{-1}Cp)$. (Proposition 9.2)
5. Prove that, given any PD^2 pair $(G, \mathcal{S})$ one can find a splitting of G over one or two compatible PD^2 pairs $(G_1, \mathcal{S}_1)$ and $(G_2, \mathcal{S}_2)$ such that G_1 and G_2 are free groups of strictly lower rank than G . Using this and the base case rank = 1, prove by induction that every such group pair encodes the fundamental data of a surface with m discs removed, where m is the number of associated subgroups in the group pair. (Theorem 9.2)
6. Returning to the setting of (4), conclude using (5) that the splitting of G over C corresponds to a gluing of two such surfaces along their removed discs. Conclude that G is a surface group. (Theorem 9.3)

2 Preliminaries

2.1 Poincaré Duality for Closed Manifolds

The definition of a Poincaré duality group is modeled on the following basic feature of closed manifolds: cohomology and homology are related by the cap product with a fundamental class. In its simplest form, if M is a closed oriented n -manifold, then there is a class

$$[M] \in H_n(M; \mathbb{Z})$$

such that, for every coefficient ring R , the cap product map

$$[M] \frown - : H^k(M; R) \rightarrow H_{n-k}(M; R)$$

is an isomorphism for all k . This is the usual statement of Poincaré duality.

An analogous statement applies for all closed manifolds. If M is not orientable, then there is no fundamental class in ordinary homology with $\mathbb{Z}$ coefficients. Instead, one uses the *orientation character*

$$w : \pi_1(M) \rightarrow \text{Aut}(\mathbb{Z}) \cong \{\pm 1\}.$$

This map records whether a loop in M preserves or reverses local orientation. This gives an action of $\pi_1(M)$ on $\mathbb{Z}$, and thereby makes $\mathbb{Z}$ into a $\mathbb{Z}\pi_1(M)$ -module, denoted $\mathbb{Z}_w$. When M is orientable, this action is trivial and $\mathbb{Z}_w$ is the usual coefficient module $\mathbb{Z}$. We can now state Poincaré duality for closed manifolds that are not necessarily orientable.

Theorem 2.1 (Twisted Poincaré duality). *Let M be a closed n -manifold with orientation module $\mathbb{Z}_w$, let $G = \pi_1(M)$, and let A be a $\mathbb{Z}G$ -module. Then there is a fundamental class $[M] \in H_n(M; \mathbb{Z}_w)$ such that, for all $k \in \mathbb{Z}$, the cap product*

$$[M] \frown (-) : H^k(M; A) \rightarrow H_{n-k}(M; A \otimes \mathbb{Z}_w)$$

is an isomorphism.

2.2 Relative Poincaré for Compact Manifolds

The definition of a duality pair (see Section 5) is modeled on the statement of Poincaré duality for compact manifolds with boundary. In the context of a compact manifold M with boundary ∂M , we have two natural cap product maps. For $z \in H_\ell(M, \partial M; \mathbb{Z})$, and for any $\mathbb{Z}G$ -module A , there are maps

$$\begin{aligned} z \frown (-) : H^k(M; A) &\rightarrow H_{\ell-k}(M, \partial M; A) \\ z \frown (-) : H^k(M, \partial M; A) &\rightarrow H_{\ell-k}(M; A). \end{aligned}$$

Additionally, the definition of the orientation module in Section 2.1 extends straightforwardly to the relative case. We can thus state a theorem of relative Poincaré duality (called *Poincaré-Lefschetz duality*) in the case of an arbitrary compact n -manifold:

Theorem 2.2 (Twisted Poincaré-Lefschetz duality). *Let M be a compact n -manifold with boundary ∂M and fundamental class $[M] \in H_n(M, \partial M; \mathbb{Z}_w)$. Then for each $\mathbb{Z}G$ -module A and for $k \in \mathbb{Z}$, the cap products*

$$\begin{aligned} [M] \frown (-) : H^k(M; A) &\rightarrow H_{n-k}(M, \partial M; A \otimes \mathbb{Z}_w) \\ [M] \frown (-) : H^k(M, \partial M; A) &\rightarrow H_{n-k}(M; A \otimes \mathbb{Z}_w) \end{aligned}$$

are isomorphisms.

2.3 Closed Surfaces

We briefly recall the classification of closed surfaces and give presentations of their fundamental groups. By the classification theorem of closed surfaces, any closed connected surface is homeomorphic to one of the following:

1. The sphere S^2 (the **closed orientable surface of genus 0**).
2. Σ_g , the connected sum of g tori for $g \geq 1$ (the **closed orientable surface of genus g**).
3. N_k , the connected sum of k copies of $\mathbb{RP}^2$ (the **closed non-orientable surface of genus k**).

For a manifold M , the group homology of $\pi_1(M)$ agrees with the homology of M when M is a $K(\pi_1(M), 1)$ complex (see Section 3.2), i.e. when M is aspherical. This is equivalent to the universal cover of M being contractible. As the connection between group homology and the homology of spaces is central to this essay, we define the following:

Definition 2.1. *A group G is a **surface group** if it is isomorphic to the fundamental group of an aspherical closed surface.*

Note that the only closed connected surfaces without this property are S^2 and $\mathbb{RP}^2$, both of which have universal cover S^2 . Recall that the fundamental groups of the closed surfaces of nonzero genus may be presented as follows:

$$\pi_1(\Sigma_g) = \left\langle x_1, y_1, \dots, x_g, y_g \mid \prod_{i=1}^g [x_i, y_i] \right\rangle, \quad \pi_1(N_k) = \left\langle z_1, \dots, z_k \mid \prod_{i=1}^k z_i^2 \right\rangle \quad (2.1)$$

Then G is a surface group if and only if it can be presented in one of the forms above, for $g \geq 1$ or $k \geq 2$.

3 Group Homology

3.1 Definition

We now recall the definition of group homology and cohomology. In the following, let G be any group.

The **integral group ring** $\mathbb{Z}G$ is the free $\mathbb{Z}$ -module generated by the elements of G , whose elements are finite formal sums $\sum_{g \in G} a(g)g$ with $a(g) \in \mathbb{Z}$. The obvious action of G on $\mathbb{Z}G$ by multiplication extends uniquely to a $\mathbb{Z}$ -bilinear product $\mathbb{Z}G \times \mathbb{Z}G \rightarrow \mathbb{Z}G$. A (left) G -module will simply mean a (left) $\mathbb{Z}G$ -module. Equivalently, a G -module is an abelian group A together with a left action of G on A by automorphisms.¹

Algebraically, group homology is defined in terms of so-called *projective resolutions*. In the following, $\mathbb{Z}$ is regarded as a G -module with trivial G -action.

Definition 3.1. Let A be a G -module. A **resolution** of A (over $\mathbb{Z}G$) is an exact sequence of G -modules

$$\cdots \rightarrow L_2 \rightarrow L_1 \rightarrow L_0 \xrightarrow{\epsilon} A \rightarrow 0.$$

If each L_i is a projective G -module, then this is called a **projective resolution** of A over $\mathbb{Z}G$. If n is such that $L_i = 0$ for $i > n$, then we say that the resolution has **length** $\leq n$.

Definition 3.2. Let M be a right G -module, let N be a left G -module, and let

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Then we define the **homology** of G with coefficients in M and the **cohomology** of G with coefficients in N by

$$H_*(G; M) = H_*(M \otimes_G P_*), \quad H^*(G; N) = H^*(\text{Hom}_G(P_*, N)).$$

Here and throughout this essay, $A \otimes_G B$ is taken to mean $A \otimes_{\mathbb{Z}G} B$ for a right G -module A and a left G -module B . Note also that this definition does not depend on the choice of projective resolution P_* .

3.2 Topological Interpretation

Definition 3.3. Let G be a group, and let X be a connected CW complex with $\pi_1(X) = G$. Then X is called an **Eilenberg-MacLane complex of type** $(G, 1)$, or a **$K(G, 1)$ complex**, if the universal cover $\tilde{X}$ of X is contractible.

This condition is equivalent to the condition that $\pi_n(X)$ is trivial for $n > 1$. The most relevant property of such spaces is that their homology is determined entirely by their fundamental group.

Proposition 3.1 (page 59 in [4]). If X is a $K(G, 1)$ complex, then for any G -module A ,

$$H_*(G; A) \cong H_*(X; \mathcal{A}) \text{ and } H^*(G; A) \cong H^*(X; \mathcal{A}),$$

where $\mathcal{A}$ is the local coefficient system associated to the G -module A .² The reason for this is that under the conditions above, the cellular chain complex $C_*(\tilde{X})$ of the universal cover of the space X is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$, where the G -module structure on $C_*(\tilde{X})$ is induced by the action of G on $\tilde{X}$ by

¹Any left G -module can be seen canonically as a right G -module via the identification $x \cdot g = g^{-1}x$. In this paper, the term G -module will refer a priori to a *left* G -module.

²For more information on local coefficients, see [11], Section 3.H.

deck transformations. It is a standard fact that a $K(G, 1)$ complex exists for every group G (see e.g. [11], Example 1B.7). It follows that group homology inherits many of the structures present in the homology of topological spaces, including operations such as the cup and cap product. Additionally, if X is a $K(G, 1)$ complex satisfying any particular homological relation (such as Poincaré duality), then $G = \pi_1(X)$ will satisfy the same relation.

Another fact that will be used liberally throughout the essay is that any two $K(G, 1)$ complexes are homotopy equivalent (see e.g. [11], Theorem 1B.8). This flexibility in the choice of a $K(G, 1)$ model for G will be important for providing a conceptual understanding of the steps of the final proof of Theorem 1.1. As more algebraic properties of G are proven (typically by showing the existence of a *splitting* of G : see Section 6), the conceptual picture of an appropriate choice of $K(G, 1)$ model will become more structured.

Example 3.1. Let $G = F_n$, the free group of rank n . Then if X is a rose with n petals (i.e. the wedge sum of n circles), we can easily see that X is a $K(G, 1)$ complex, and can thus quickly compute that

$$H^i(G; \mathbb{Z}) \cong H_i(G; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & i = 0 \\ \mathbb{Z}^n & i = 1 \\ 0 & \text{otherwise.} \end{cases}$$

3.3 Dimension and Finiteness Properties

We now record several algebraic notions which will be used throughout the proof. These generally concern the existence of certain types of projective resolutions.

Definition 3.4. A group G is of **type FP_n** ($n \geq 0$) if there exists a partial projective resolution

$$P_n \rightarrow \cdots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of $\mathbb{Z}$ over $\mathbb{Z}G$ such that each P_i is finitely generated. A group is of **type FP** if there exists a projective resolution P_* of $\mathbb{Z}$ over $\mathbb{Z}G$ of finite length such that each P_i is finitely generated.

It is easy to see that a group is of type FP_1 if and only if it is finitely generated. It will also be useful to note that if a finitely generated group is presented as $G = \langle S \mid R \rangle$ with S finite, then G is of type FP_2 if and only if N_{ab} is finitely generated as a G -module, where N is the normal closure of R in the free group F_S (see [4], Exercise VIII.5.3). It is also clear that a group of type FP is of type FP_n for all $n \geq 0$.³

A related and important notion is that of cohomological dimension.

Definition 3.5. The **cohomological dimension** of a group G , denoted $\text{cd } G$, is defined by

$$\text{cd } G = \sup\{n \mid H^n(G; M) \neq 0 \text{ for some } G\text{-module } M\}.$$

Equivalently, $\text{cd } G$ is the minimal (possibly infinite) length of a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$.⁴

The following deep theorem, proven in a restricted case by Stallings in [16] and in full generality by Swan in [18], will be of use frequently throughout this essay:

Theorem 3.1 (Stallings–Swan). *A group is of cohomological dimension 1 if and only if it is free.*

³Note that none of these conditions imply that G is finitely presented. In fact, groups of type FP that are not finitely presented form important counterexamples in the theory of Poincaré duality groups, as is mentioned in Section 10.

⁴See [4], Lemma VIII.2.1.

Finally, the following standard fact will greatly simplify certain parts of the proof:

Proposition 3.2 (Corollary VIII.2.5 in [4]). *If a group G is such that $\text{cd } G < \infty$, then G is torsion-free.*

This follows from the fact that nontrivial finite cyclic groups have infinite cohomological dimension, and that $\text{cd } H \leq \text{cd } G$ whenever $H \leq G$.

4 Duality Groups

4.1 Poincaré Duality Groups

Recall from Section 3.2 that the isomorphisms $H_*(G; A) \cong H_*(X; \mathcal{A})$ and $H^*(G; A) \cong H^*(X; \mathcal{A})$ for a $K(G, 1)$ complex X and any G -module A induce cap product maps

$$\frown: H_n(G; B) \otimes H^k(G; A) \rightarrow H_{n-k}(G; B \otimes A)$$

for all $n, k \in \mathbb{Z}$ and all G -modules A, B . We can therefore formulate Poincaré duality as a property of abstract groups.

Definition 4.1. *A group G is called a **Poincaré duality group** of dimension n if there exists a $\mathbb{Z}G$ -module $\tilde{\mathbb{Z}}$, isomorphic to $\mathbb{Z}$ as an abelian group, and an element $e \in H_n(G; \tilde{\mathbb{Z}})$, such that the map*

$$[e] \frown (-) : H^k(G; A) \rightarrow H_{n-k}(G; \tilde{\mathbb{Z}} \otimes A)$$

*is an isomorphism for all integers k and all G -modules A . The G -module $\tilde{\mathbb{Z}}$ is called the **dualizing module**, and e is called a **fundamental class** of G .*

This definition aligns closely with the formulation of Theorem 2.1. Indeed, the orientation module $\mathbb{Z}_w$ was defined for a manifold M with $G = \pi_1(M)$ such that

- $\mathbb{Z}_w$ has trivial G -action if and only if M is orientable.⁵
- In any case, $\mathbb{Z}_w$ is isomorphic as an abelian group to $\mathbb{Z}$.

Theorem 2.2 and Proposition 3.1 tell us that the fundamental group of any closed aspherical n -manifold is a PD^n group. Such groups are, of course, the prototypical examples of Poincaré duality groups; it is not at all easy to find a Poincaré duality group that is not of this form.⁶

4.2 Duality Groups

Whereas Poincaré duality groups mimic the homological behavior of closed manifolds, the more general concept of a duality group encompasses groups that mimic (under certain conditions) the homological behavior of compact aspherical manifolds with boundary.⁷ This weaker version of the PD^n condition was first formulated by Bieri and Eckmann in [1], and is central to the proof of Theorem 1.1.

Definition 4.2. *Let G be a group. Then G is a **duality group** of dimension n , or a **D^n group**, if there exists a $\mathbb{Z}G$ -module C and an element $e \in H_n(G; C)$ such that the cap product map*

$$e \frown (-) : H^k(G; A) \rightarrow H_{n-k}(G; C \otimes A)$$

*is an isomorphism for all integers k and all G -modules A . The G -module C is called the **dualizing module**, and e is called a **fundamental class** of G .*

⁵In analogy with this fact, we say that a PD^n group G is *orientable* if the dualizing module $\tilde{\mathbb{Z}}$ has trivial G -action, and that G is *non-orientable* otherwise.

⁶See Section 10 for further discussion of this.

⁷This relationship is made explicit in Proposition 5.3.

We now state some of the algebraic properties of duality groups that are relevant to this essay.

Proposition 4.1. *Let G be a D^n group with dualizing module C and fundamental class e . Then*

1. $C \cong H^n(G; \mathbb{Z}G)$, with $C \neq 0$.
2. $\text{cd } G = n$.
3. $H_n(G; C)$ is infinite cyclic generated by e .

Proof. Taking $k = n$ and $A = \mathbb{Z}G$ in the duality isomorphism gives that

$$H^n(G; \mathbb{Z}G) \cong H_0(G; C \otimes \mathbb{Z}G) = (C \otimes \mathbb{Z}G) \otimes_G \mathbb{Z} \cong C \otimes_G \mathbb{Z}G = C.$$

Taking $k = 0$ and $A = \mathbb{Z}$, we also get that $H_n(G; C) \cong H^0(G; \mathbb{Z}) \cong \mathbb{Z}$, and since $e \frown 1 = e$, $H_n(G; C)$ is generated by e . This shows that $C \neq 0$ and that $\text{cd } G \geq n$. Since $H^k(G; A) = H_{n-k}(G; C \otimes A) = 0$ for $k > n$, we get that $\text{cd } G \leq n$. Thus $\text{cd } G = n$. $\square$

As an immediate corollary to this and Proposition 3.2, we have the following result:

Corollary 4.1. *All duality groups are torsion-free.*

Duality groups satisfy strong finiteness properties:

Proposition 4.2. *All duality groups are of type FP.*

Proof. By duality, and since the homology functor preserves direct limits (see [4], p. 121), the cohomology functor preserves direct limits in any duality group. Then Brown's finiteness condition (Theorem 3 in [5]) gives the result. $\square$

We can now rather quickly prove the following important result, which fully classifies duality groups of dimension one:

Proposition 4.3 (Dimension 1). *A group is a D^1 group if and only if it is free of finite rank. A group is a PD^1 group if and only if it is infinite cyclic.*

Proof. If G is a duality group of dimension 1, then G has cohomological dimension 1, so the Stallings–Swan theorem (Theorem 3.1) shows that G is free.

Indeed, a rose with r petals is a $K(F_r, 1)$, so its universal cover gives a length-one free resolution of $\mathbb{Z}$ over $\mathbb{Z}F_r$. A direct computation from this resolution shows that F_r is a D^1 group with dualizing module $H^1(F_r; \mathbb{Z}F_r)$.⁸ The dualizing module $H^1(F_r; \mathbb{Z}F_r)$ is infinite cyclic as an abelian group exactly when $r = 1$. Thus among finite-rank free groups, the only PD^1 group is $F_1 \cong \mathbb{Z}$. $\square$

The following theorem of Strebel, proved in [17], will be used much later in the proof.

Theorem 4.1. *Let G be a PD^n group, and let L be a subgroup of G of infinite index. Then L has cohomological dimension at most $n - 1$.*

From this, we have the following corollary:

Corollary 4.2. *Let G be a PD^2 group, and let L be a nontrivial subgroup of G of infinite index. Then L is a free group.*

Proof. Theorem 4.1 gives that L has cohomological dimension at most 1; as L is nontrivial, we must have that $\text{cd } L = 1$. Theorem 3.1 therefore gives that L is free. $\square$

⁸Alternatively, see Proposition 5.1 of [1].

5 Group Pairs

The proof of Theorem 1.1 involves splitting a PD^2 group along various subgroups. It is therefore useful to be able to simultaneously record the data of a larger group and a set of associated subgroups. This motivates the notion of a *group pair*, and especially of a *duality pair*. Duality pairs were first introduced and studied by Bieri and Eckmann in [2], and the definitions and proofs in this section are largely adapted from that paper.

5.1 Relative Homology and Cap Products

Definition 5.1. Let G be a group, and let $\mathcal{S} = \{S_i\}_{i \in I}$ be a family of subgroups of G . Then the tuple $(G, \mathcal{S})$ is called a **group pair**.

For a group pair $(G, \mathcal{S})$ with $\mathcal{S} \neq \emptyset$, write $\mathbb{Z}G/\mathcal{S}$ to denote the left G -module

$$\bigoplus_{i \in I} \mathbb{Z}G \otimes_{S_i} \mathbb{Z} = \bigoplus_i \mathbb{Z}G/S_i.$$

This is the free abelian group on the set of left cosets of S_i in G , with G acting by left multiplication on each summand. Then we define the **augmentation homomorphism** $\epsilon : \mathbb{Z}G/\mathcal{S} \rightarrow \mathbb{Z}$ by $\epsilon(hS_i) = 1$ for all left cosets hS_i , and the kernel of this map is denoted $\Delta = \Delta_{G/\mathcal{S}}$.

We may now define a relative version of homology for group pairs:

Definition 5.2. Let $(G, \mathcal{S})$ be a group pair. If $\mathcal{S} \neq \emptyset$, then we define

$$H_k(G, \mathcal{S}; A) = H_{k-1}(G; \Delta \otimes A), \quad H^k(G, \mathcal{S}; A) = H^{k-1}(G; \text{Hom}(\Delta, A)),$$

where G acts diagonally on both $\text{Hom}(\Delta, A)$ and on $\Delta \otimes A$. If $\mathcal{S} = \emptyset$, then we define $H_k(G, \mathcal{S}; A) = H_k(G; A)$, $H^k(G, \mathcal{S}; A) = H^k(G; A)$.

Just as in the case of the relative homology of manifolds with boundary, there are two natural cap products relating the absolute homology groups of a group G and the relative homology groups of a group pair $(G, \mathcal{S})$. The following proposition gives a general way to define a cap product map in a variety of settings:

Proposition 5.1 (See (2.1) in [2]). Let G be a group, let A, C , and D be G -modules, and let $\mathfrak{B} : C \otimes A \rightarrow D$ be a G -homomorphism. Then for any right G -module C , and for any $n, k \in \mathbb{Z}$, there is a natural cap product map

$$(-) \frown (-) : H_n(G; C) \otimes H^k(G; A) \rightarrow H_{n-k}(G; D).$$

Definition 5.3. Suppose that $(G, \mathcal{S})$ is a group pair with $\mathcal{S} \neq \emptyset$. Then the **identity cap product** on $(G, \mathcal{S})$ is the cap product

$$(-) \frown (-) : H_{r+s}(G, \mathcal{S}; C) \otimes H^r(G; A) \rightarrow H_s(G, \mathcal{S}; C \otimes A)$$

corresponding via Proposition 5.1 to the identity G -homomorphism $(\Delta \otimes C) \otimes A \rightarrow \Delta \otimes C \otimes A$. The **evaluation cap product** on $(G, \mathcal{S})$ is the cap product

$$(-) \frown (-) : H_{r+s}(G, \mathcal{S}; C) \otimes H^r(G, \mathcal{S}; A) \rightarrow H_s(G; C \otimes A)$$

corresponding to the “evaluation” G -homomorphism $\mathfrak{B} : \Delta \otimes C \otimes \text{Hom}(\Delta, A) \rightarrow C \otimes A$ defined by $\mathfrak{B}(d \otimes c \otimes h) = c \otimes h(d)$.

For a family of groups $\mathcal{S} = \{S_i : i \in I\}$, we write

$$H_k(\mathcal{S}; A) = \oplus_i H_k(S_i; A) \text{ and } H^k(\mathcal{S}; A) = \prod_i H^k(S_i; A),$$

where A is regarded as an S_i -module for each $i \in I$ by restriction. The relationship between the two cap products of Definition 5.3, and between relative and absolute group homology, is encapsulated by the following “cap product ladder”, which will be of use throughout the proof of Theorem 1.1:

Theorem 5.1 (Theorem 2.1 in [2]). *Let $(G, \mathcal{S})$ be a group pair with $\mathcal{S} \neq \emptyset$, and let A and C be G -modules, with $e \in H_n(G, \mathcal{S}; C)$. Then the following diagram commutes, and the rows are exact:*

$$\begin{array}{ccccccc} \cdots & \xrightarrow{p^*} & H^k(G; A) & \xrightarrow{\text{res}} & H^k(\mathcal{S}; A) & \xrightarrow{\delta} & H^{k+1}(G, \mathcal{S}; A) \xrightarrow{p^*} \cdots \\ & & \downarrow e \frown (-) & & \downarrow \partial e \frown (-) & & \downarrow e \frown (-) \\ \cdots & \xrightarrow{p^*} & H_{n-k}(G, \mathcal{S}; C \otimes A) & \xrightarrow{\partial} & H_{n-k-1}(\mathcal{S}; C \otimes A) & \xrightarrow{\text{cor}} & H_{n-k-1}(G; C \otimes A) \xrightarrow{p^*} \cdots \end{array}$$

Here the rows are the long exact cohomology and homology sequences of the pair $(G, \mathcal{S})$: res and cor are induced by the inclusions $S_i \hookrightarrow G$, and δ and ∂ are the connecting homomorphisms.

We may now formulate a relative version of Poincaré duality for group pairs, in a form that closely mirrors Theorem 2.2.

Definition 5.4. *A group pair $(G, \mathcal{S})$ is a **duality pair** of dimension $n \geq 0$, or a **Dⁿ pair**, if there exists a dualizing G -module C , and a fundamental class $e \in H_n(G, \mathcal{S}; C)$, such that the cap products*

$$\begin{aligned} e \frown (-) : H^k(G; A) &\rightarrow H_{n-k}(G, \mathcal{S}; C \otimes A) \\ e \frown (-) : H^k(G, \mathcal{S}; A) &\rightarrow H_{n-k}(G; C \otimes A) \end{aligned}$$

*are isomorphisms for all G -modules A and all integers k .⁹ $(G, \mathcal{S})$ is called a **Poincaré duality pair** of dimension n , or a **PDⁿ pair**, if C is isomorphic to $\mathbb{Z}$ as an abelian group.*

We now list and prove the relevant properties of duality pairs. It will be shown that duality pairs have a rather fixed structure, and that in particular the structure of Poincaré duality pairs of dimension 2 is very rigid. We start with the basic consequences of the definition:

Proposition 5.2. *Let $(G, \mathcal{S})$ be a D^n pair. Then*

1. C is determined by the pair $(G, \mathcal{S})$.
2. $H_n(G, \mathcal{S}; C) = \mathbb{Z}$, generated by the fundamental class e .
3. $n = \text{cd}(G, \mathcal{S})$, and so $(G, \mathcal{S})$ is not a D^m pair for $m \neq n$.

Proof. $H^n(G, \mathcal{S}; \mathbb{Z}G) \cong H_0(G; C \otimes \mathbb{Z}G) \cong \mathbb{Z} \otimes_{\mathbb{Z}G} (C \otimes_{\mathbb{Z}} \mathbb{Z}G) \cong \mathbb{Z} \otimes_{\mathbb{Z}G} C \cong C$, which proves (1). $H_n(G, \mathcal{S}; C) \cong H^0(G; \mathbb{Z}) = \mathbb{Z}$ by duality, and since $e \frown 1 = e$, $H_n(G, \mathcal{S}; C)$ is generated by e . Duality gives that $H^k(G, \mathcal{S}; A) = 0$ for all G -modules A and all $k > n$. $H^n(G, \mathcal{S}; \mathbb{Z}G) = C \neq 0$, so $\text{cd}(G, \mathcal{S}) = n$, and thus $(G, \mathcal{S})$ cannot be a D^m pair for $m \neq n$. $\square$

⁹Note that in the case $\mathcal{S} = \emptyset$, this is equivalent to the group G being a duality group in the sense of Definition 4.2.

The following proposition gives legitimacy to the idea that duality groups behave like manifolds with nonempty boundary, and that a duality pair carries the data of the manifold and its boundary components.¹⁰

Proposition 5.3 (Theorem 4.2 in [2]). *Let $(G, \mathcal{S})$, $\mathcal{S} = \{S_i \mid i \in I\}$, be a D^n pair with dualizing module C . Then we have the following:*

1. $\mathcal{S}$ is a finite family of subgroups of G .
2. G is a duality group of dimension $n - 1$ with dualizing module $\Delta \otimes C$.
3. Each S_i is a duality group of dimension $n - 1$ with dualizing module C .
4. A choice of fundamental class $e \in H_n(G, \mathcal{S}; C)$ determines a choice of fundamental class $e_i \in H_{n-1}(S_i; C)$ for each $S_i \in \mathcal{S}$. This choice is given by $\partial e = \sum_i e_i$, where ∂ is the connecting homomorphism in the exact homology sequence of G modulo $\mathcal{S}$.

Proof. By definition, e is an element of $H_{n-1}(G; \Delta \otimes C)$, and $(e \smallfrown -) : H^k(G; A) \rightarrow H_{n-k-1}(G; \Delta \otimes C \otimes A)$ is an isomorphism for all A and k , so G is a D^{n-1} group with dualizing module $\Delta \otimes C$, proving (2). By Theorem 5.1, we have a commutative diagram

$$\begin{array}{ccccccc}
 \cdots & \xrightarrow{p^*} & H^k(G; A) & \xrightarrow{\text{res}} & H^k(\mathcal{S}; A) & \xrightarrow{\delta} & H^{k+1}(G, \mathcal{S}; A) \xrightarrow{p^*} \cdots \\
 & & \downarrow e \smallfrown (-) & & \downarrow \partial e \smallfrown (-) & & \downarrow e \smallfrown (-) \\
 \cdots & \xrightarrow{p^*} & H_{n-k}(G, \mathcal{S}; C \otimes A) & \xrightarrow{\partial} & H_{n-k-1}(\mathcal{S}; C \otimes A) & \xrightarrow{\text{cor}} & H_{n-k-1}(G; C \otimes A) \xrightarrow{p^*} \cdots
 \end{array}$$

Since both maps $e \smallfrown (-)$ are isomorphisms, the 5-lemma gives that $\partial e \smallfrown (-)$ is also an isomorphism. $\partial e \in H_{n-1}(\mathcal{S}; C) = \oplus_i H_{n-1}(S_i; C)$ decomposes uniquely as $\sum_i e_i$ for some elements $e_i \in H_{n-1}(S_i; C)$, and thus the cap product $\partial e \smallfrown (-) : \prod_i H^k(S_i; A) \rightarrow \oplus_i H_{n-k-1}(S_i; C \otimes A)$ decomposes as a product of cap products $e_i \smallfrown (-) : H^k(S_i; A) \rightarrow H_{n-k-1}(S_i; C \otimes A)$ over $i \in I$. Since the map $\partial e \smallfrown (-)$ is an isomorphism, the family must be finite, and each $e_i \smallfrown (-)$ must be an isomorphism for all G -modules A and all $k \in \mathbb{Z}$.

To prove (iii) and (iv), we need to show that the maps $e_i \smallfrown (-)$ are isomorphisms for all i and for all S_i -modules B . For such a B , let A denote the G -module $\mathbb{Z}G \otimes_{S_i} B$, and note that

$$A = (\mathbb{Z}S_i \oplus \mathbb{Z}[G \setminus S_i]) \otimes_{S_i} B \cong B \oplus (\mathbb{Z}[G \setminus S_i] \otimes_{S_i} B).$$

Since the maps $e_i \smallfrown (-)$ are isomorphisms with respect to A , they must be isomorphisms with respect to both summands, so in particular with respect to B . This completes the proof. $\square$

Finally, we turn to the specific case of Poincaré duality pairs in dimension 2. In this setting, the very rigid structure of D^1 groups (Proposition 4.3) induces a similarly rigid structure on PD^2 pairs.

Theorem 5.2. *Let $(G, \mathcal{S})$ be a PD^2 pair with $\mathcal{S} \neq \emptyset$. Then G is a free group of finite rank, and $\mathcal{S}$ is a finite family of infinite cyclic subgroups of G .*

Proof. From Proposition 5.3 (1), we know that the family $\mathcal{S}$ is finite. From Proposition 5.3 (2), we see that G is a D^1 group, and thus G is free of finite rank by Proposition 4.3. Similarly, Proposition 5.3 (3) gives that each $S \in \mathcal{S}$ is a D^1 group with dualizing module $\mathbb{Z}$, i.e. a PD^1 group, and is therefore infinite cyclic (again by Proposition 4.3). $\square$

¹⁰Indeed, Proposition 5.3 gives the following immediate corollary: if M is a compact aspherical n -manifold with finitely many boundary components $\partial_1 M, \dots, \partial_k M$ such that each $\partial_i M$ is aspherical and each map $\pi_1(\partial_i M) \rightarrow \pi_1(M)$ is injective, then $(\pi_1(M); \{\pi_1(\partial_i M)\}_{i=1}^k)$ is a D^n pair and $\pi_1(M)$ is a duality group of dimension $n - 1$.

The proof of Theorem 1.1 will also rely crucially on the following two results, both of which state in essence that the property of being a PD^2 pair is stable under appropriate “gluing and splitting” operations. For more detail on amalgamated products, HNN extensions, and graphs of groups, see Section 6.2 below.

Proposition 5.4 (Theorem 8.1 in [2]). *Let $(G_\alpha; \mathcal{S}_\alpha)$ be group pairs for $\alpha = 1, 2$, with subgroups $T \leq G_1$ and $T' \leq G_2$ such that $T \cong T'$. Let $(G; \mathcal{S})$ be the pair consisting of $G = G_1 *_T G_2$ (the amalgamated product over $T \cong T'$) and $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$. Then we have the following:*

1. *If $(G_1; \mathcal{S}_1 \cup \{T\})$ and $(G_2; \mathcal{S}_2 \cup \{T'\})$ are PD^n pairs, then so is $(G, \mathcal{S})$.*
2. *If $(G; \mathcal{S})$ is a PD^n pair and T is a PD^{n-1} group, then $(G_1; \mathcal{S}_1 \cup \{T\})$ and $(G_2; \mathcal{S}_2 \cup \{T'\})$ are PD^n pairs.*

Proposition 5.5 (Theorem 8.3 in [2]). *Let $(G_1; \mathcal{S})$ be a group pair, with $T_1, T_2 \subset G_1$ such that $T_1 \cong T_2$. Let $(G; \mathcal{S})$ be the pair consisting of $G = G_1 *_{T_1 \sim T_2}$ and the same associated subgroups $\mathcal{S}$. Then we have the following:*

1. *If $(G_1; \mathcal{S} \cup \{T_1, T_2\})$ is a PD^n pair, then so is $(G, \mathcal{S})$.*
2. *If $(G; \mathcal{S})$ is a PD^n pair and T_1 is a PD^{n-1} group, then $(G_1; \mathcal{S} \cup \{T_1, T_2\})$ is a PD^n pair.*

The proofs of Propositions 5.4 and 5.5 in [2] use the Mayer–Vietoris sequences for the relative homology of group pairs associated to amalgamated products and HNN extensions, together with the compatibility of these sequences with the cap-product ladder of Theorem 5.1.

6 Trees and Bass–Serre Theory

The next major tool used in the proof of Theorem 1.1 is *Bass–Serre theory*, which establishes a certain equivalence between algebraic splittings of a group and actions of that group on a tree. In this way, the problem of finding nontrivial splittings of a group G (the goal of large parts of the proof of Theorem 1.1) is reduced to the problem of finding a suitable action of G on a tree.

This section thus has two purposes. First, we state and prove a criterion which identifies certain partially ordered sets with the oriented edge sets of trees. This will be used in Section 8, where a collection of subsets of G will be converted into a tree. Second, we recall the one-edge form of the main theorem of Bass–Serre theory, which is the form used throughout the proof of Theorem 1.1.

6.1 Trees and Partial Orders

First we describe the conventions used in this essay (and in [13]) for graphs. A **graph** Γ consists of two sets $E = E(\Gamma)$ and $V = V(\Gamma)$, a function $E \rightarrow E$ sending $e \in E$ to $\bar{e}$, and a map $\iota : E \rightarrow V$, such that $\bar{\bar{e}} = e$ and $\bar{e} \neq e$. Given a graph Γ , we also have a map $\tau : E \rightarrow V$ defined by $\tau(e) = \iota(\bar{e})$.

An element $e \in E(\Gamma)$ then represents a single edge in the graph Γ along with an orientation of that edge, such that e is said to have **initial vertex** $\iota(e)$ and **terminal vertex** $\tau(e)$. An unoriented edge of Γ is thus represented by an **edge pair** $(e, \bar{e})$. For $u, v \in V(\Gamma)$, an **edge path** from u to v is an n -tuple $e_1 e_2 \dots e_n$ of oriented edges such that $u = \iota(e_1)$, $v = \tau(e_n)$, and $\tau(e_i) = \iota(e_{i+1})$ for $1 \leq i < n$. A graph Γ is **connected** if every pair of points in Γ is connected by some path. A path is **reduced** if it contains no consecutive pair of the form $e\bar{e}$. A connected graph Γ is a **tree** if the only reduced path from any vertex $v \in V(\Gamma)$ to itself is the trivial path. Finally, note that a tree can be made into a 1-dimensional CW complex

in the obvious way; an action of G on a tree will mean a cellular action on this associated CW complex.

The following characterization of trees as partial orders satisfying a certain condition (from [8], Theorem 2.1) is central to the proof of the main theorem.

Lemma 6.1. *Let X be a tree. Then there exists a partial ordering on the edge set $E(X)$ such that $e \preceq f$ if and only if there exists a reduced edge path $e_1, \dots, e_n$ with $e_1 = e$ and $e_n = f$.¹¹ This ordering has the following properties:*

1. *If $e \preceq f$, then $\bar{f} \preceq \bar{e}$.*
2. *If $e \preceq f$, then there are only finitely many g with $e \preceq g \preceq f$.*
3. *For any pair e, f , at least one of $e \preceq f$, $e \preceq \bar{f}$, $\bar{e} \preceq f$, $\bar{e} \preceq \bar{f}$ holds.*
4. *For no pair e, f is $e \preceq f$ and $e \preceq \bar{f}$.*

Conversely, if $(\epsilon, \preceq, -)$ is a partially ordered set with an involution $e \rightarrow \bar{e}$, $e \mapsto \bar{e}$, $\bar{\bar{e}} = e$, such that the four conditions above are satisfied, then there exists a tree X with $E(X) = \epsilon$ such that the order relation $\preceq$ is the same as the edge-path order relation on $E(X)$ and such that $\bar{e}$ is the inverse edge of e .

Proof. The fact that the properties (1)-(4) are satisfied by the ordering $\preceq$ on any tree X is easy to check. (1) follows by reversing the reduced edge path from e to f . (3) follows from the fact that X is connected, and thus there must exist a reduced edge path between any two unoriented edges of X , and (2) follows from the fact that this unoriented edge path is unique in a tree (and always of finite length). If there existed a pair violating (4), it would produce two distinct edge paths between the unoriented edges represented by e and f , which is impossible in a tree, so (4) holds too.

Suppose now that $(\epsilon, \preceq, -)$ satisfies conditions (1)-(4). First, for $e, f \in \epsilon$, write that $e \leq f$ if $e \preceq f$, $e \neq f$, and for $g \in \epsilon$, $e \preceq g \preceq f$ implies that $g = e$ or $g = f$. Define a relation $\sim$ on ϵ by $e \sim f$ if $e = f$ or $e \leq \bar{f}$. This relation is reflexive, and is symmetric by (1).

Now suppose $e \sim f$ and $f \sim h$. If $e = f$, $e = h$, or $f = h$, then certainly $e \sim h$. Thus we can suppose that $e \leq \bar{f}$ and $f \leq \bar{h}$, with $e \neq h$. We now use (3), and rule out the other three options, leaving only $e \preceq \bar{h}$: first, if $e \preceq h$, then $e \preceq h \preceq \bar{f}$, which contradicts $e \leq \bar{f}$ since $h \neq e$ and $h \neq \bar{f}$. If $\bar{e} \preceq \bar{h}$, then by (1) we see that $f \preceq \bar{e} \preceq \bar{h}$, which contradicts $f \leq \bar{h}$ since $\bar{e} \neq f$ and $\bar{e} \neq \bar{h}$. Finally, if $\bar{e} \preceq h$, then by (1) we have $\bar{h} \preceq e \preceq \bar{f}$, so $f \preceq h$, contradicting (4).

Thus $e \preceq \bar{h}$ by (3), and indeed $e \neq \bar{h}$ since $\bar{e} \not\preceq h$. Now let g be such that $e \preceq g \preceq \bar{h}$. If $\bar{f} \preceq g$, then $g \preceq \bar{h}$ implies $\bar{f} \preceq \bar{h}$, contradicting (4) since $f \preceq \bar{h}$. The case when $\bar{f} \preceq \bar{g}$ is similar. By (3), then, either $f \preceq g$ or $f \preceq \bar{g}$. If $f \preceq g$, then $f \preceq g \preceq \bar{h}$ implies $g = \bar{h}$ since $f \leq \bar{h}$. Otherwise, $e \preceq g \preceq \bar{f}$ implies $g = e$ since $e \leq \bar{f}$. Thus $g = e$ or $g = \bar{h}$, so $e \leq \bar{h}$ and the relation $\sim$ is transitive. Therefore $\sim$ is an equivalence relation.

Let Γ be the graph defined by $E(\Gamma) = \epsilon$ and $V(\Gamma) = \epsilon / \sim$, with $\iota(e) = [e]_\sim$ and $\tau(e) = [\bar{e}]_\sim$. Notice that $\tau(e) = \iota(f)$ if and only if $e \leq f$ or $e = \bar{f}$. From (2) it is easily seen that $e \preceq f$ if and only if there exists an edge path connecting e and f , and so Γ is a connected graph by (3). If there were a circuit in Γ , it would imply the existence of $[e]_\sim, [f]_\sim \in V(\Gamma)$ such that $e \preceq f \preceq e$ with $e \neq f$, which is impossible since $\preceq$ is a partial order. Therefore Γ is a tree with induced order relation $\preceq$. $\square$

¹¹Recall that elements of $E(X)$ are *oriented* edges; this condition thus corresponds to the ability to traverse X from e to f in a single direction fixed by e and f .

6.2 Splittings and Bass–Serre Theory

Definition 6.1. A group G is said to **split nontrivially** over a subgroup $C \leq G$ if G can be written as an amalgamated product $A *_C B$ or as an HNN extension $A *_C$ for some subgroups $A, B, C \leq G$ such that $A \neq C$ and $B \neq C$.¹²

Geometrically, an amalgamated product corresponds to a gluing of two spaces along a common subspace, whereas an HNN extension corresponds to gluing a space onto itself. This description motivates the following theorem, which is a special case of the main theorem of Bass–Serre theory:

Theorem 6.1. Let G be a group acting without edge inversions¹³ on a tree X . Let Y denote the quotient graph $G \backslash X$, and suppose that Y has only one edge pair $(e, \bar{e})$.

1. If $|V(Y)| = 2$, then

$$G \cong \text{stab}_G(\tilde{v}_1) *_{\text{stab}_G(\bar{e})} \text{stab}_G(\tilde{v}_2),$$

where $\tilde{e}$ is a lift of e in X and $\tilde{v}_1$ and $\tilde{v}_2$ are lifts in X of $\iota(e)$ and $\tau(e)$ respectively.

2. If $|V(Y)| = 1$, then

$$G \cong \text{stab}_G(\tilde{v}) *_{\text{stab}_G(\bar{e})},$$

where $\tilde{e}$ and $\tilde{v}$ are lifts in X of e and $\iota(e)$ respectively.

7 Almost Invariant Sets and Stallings' Theorem

In the previous section, we saw that a partial order satisfying certain conditions is equivalent to a tree, and that an action of a group on a tree gives an algebraic decomposition of the group. The missing ingredient is to construct such a partial order; this section provides the background necessary to understand this construction as it is developed in Section 8.

The key mathematical objects involved are *almost invariant sets*, which in a certain sense act like one side of a splitting of a group. Given a group G , its almost invariant subsets can be studied cohomologically through the group $H^1(G; \mathbb{Z}_2 G)$. The proof of Theorem 8.1 converts a certain family of such sets into a tree, and Theorem 6.1 then produces a splitting of a group. This is the primary mechanism behind the proofs of both Theorem 8.1 and Stallings' theorem on ends (see Section 7.3).

7.1 Almost Invariant Sets and Ends

Definition 7.1. For two sets A, B , we say that A and B are **almost equal**, and write that $A \stackrel{a}{=} B$, if A and B differ by finitely many elements. For a group G and any subset A of G , the set $\{g \mid Ag \stackrel{a}{=} A\}$ is a subgroup of G , and so we may say that A is **almost invariant** if $Ax \stackrel{a}{=} A$ for all x in some (any) generating set of G .

Denote by $\overline{\mathbb{Z}_2 G}$ the G -module $\text{Hom}(\mathbb{Z}G, \mathbb{Z}_2)$ with G -action

$$fx(z) = f(zx^{-1}), \quad yf(z) = f(y^{-1}z)$$

for $f : \mathbb{Z}G \rightarrow \mathbb{Z}_2$ and $x, y \in G$.

¹²For further information on amalgamated products and HNN extensions in the context of Bass–Serre theory, as well as a proof of Theorem 6.1, see e.g. Chapter 1 of [14].

¹³This means that $ge \neq \bar{e}$ for any $e \in E(X)$.

We have an embedding $\iota : \mathbb{Z}_2 G \rightarrow \overline{\mathbb{Z}_2 G}$ sending $x \in G$ to the characteristic function $\chi_{\{x\}} : \mathbb{Z} G \rightarrow \mathbb{Z}_2$ defined by $\chi_{\{x\}}(z) = 1$ if and only if $z = x$ (for $z \in G$). Denote by ϵG the quotient module $\overline{\mathbb{Z}_2 G} / \mathbb{Z}_2 G$ with canonical projection $\text{pr} : \overline{\mathbb{Z}_2 G} \rightarrow \epsilon G$.

It is easy to see that $\mathbb{Z}_2 G$ is identified with the set of finite subsets of G , and that $\overline{\mathbb{Z}_2 G}$ is identified with the set of subsets of G , and thus that ϵG is identified (via $[A] \mapsto [\chi_A]$) with the set of equivalence classes of subsets of G up to $\overset{a}{=}$ -equivalence. Indeed, a subset A of G is almost invariant if and only if its image in ϵG is invariant (i.e. $Ax = A$ for all $x \in \epsilon G$), or equivalently if $[\chi_A] \in (\epsilon G)^G = H^0(G; \epsilon G)$. Note that $\overline{\mathbb{Z}_2 G}$ is a co-induced module, and is thus H^* -acyclic by Shapiro's Lemma (Proposition III.6.2 in [4]). We have an exact sequence

$$0 \rightarrow \mathbb{Z}_2 G \xrightarrow{\iota} \overline{\mathbb{Z}_2 G} \xrightarrow{\text{pr}} \epsilon G \rightarrow 0$$

which induces the long exact sequence in cohomology

$$0 \rightarrow H^0(G; \mathbb{Z}_2 G) \rightarrow H^0(G; \overline{\mathbb{Z}_2 G}) \rightarrow H^0(G; \epsilon G) \xrightarrow{\delta} H^1(G; \mathbb{Z}_2 G) \rightarrow H^1(G; \overline{\mathbb{Z}_2 G}) = 0.$$

Now, $H^0(G; \mathbb{Z}_2 G) = 0$ since $(\mathbb{Z}_2 G)^G = \{0\}$, and since $\emptyset$ and G are the only invariant subsets of G , $H^0(G; \overline{\mathbb{Z}_2 G}) = \mathbb{Z}_2$ (with elements 0 and χ_G), so we have a short exact sequence (of $\mathbb{Z}_2$ -vector spaces)

$$0 \rightarrow \mathbb{Z}_2 \rightarrow H^0(G; \epsilon G) \xrightarrow{\delta} H^1(G; \mathbb{Z}_2 G) \rightarrow 0,$$

so that $\dim_{\mathbb{Z}_2} H^0(G; \epsilon G) = 1 + \dim_{\mathbb{Z}_2} H^1(G; \mathbb{Z}_2 G)$. When G is finitely generated, we call this number the number of **ends** of G .

Finally, for an arbitrary subset $U \subseteq G$, denote by $\langle U \rangle$ the element $\delta \text{pr}(\chi_U) \in H^1(G; \mathbb{Z}_2 G)$, and say that U is **trivial** if $\langle U \rangle = 0$, i.e. if $U \overset{a}{=} \emptyset$ or $U \overset{a}{=} G$.

We end this subsection with a lemma that helps turn almost equalities into equalities, up to a finite difference of sets.

Definition 7.2. Let G be a group with $T \leq G$. Every subset U of G can be written in the form $U = \bigsqcup_{i \in I} U_i x_i$, where $U_i = U x_i^{-1} \cap T \subseteq T$. The sets U_i are called the **T -components** of U . U is called **T -canonical** if whenever a component U_i of U is trivial with respect to T (in the sense that $U_i \overset{a}{=} T$ or $U_i \overset{a}{=} \emptyset$), then it is actually equal to $\emptyset$ or to T .

Recall that a group is called **locally finite** if all of its finitely generated subgroups are finite.

Lemma 7.1 (Lemma 5.1. in [2]). Let T be a subgroup of a group G that is not locally finite. Then all almost T -invariant subsets of G are almost T -canonical.

Proof. Let U be an almost T -invariant subset of G . Choose C to be a finite subset of T , closed under inverses, which generates an infinite subgroup T' of T . Let $\{U_i\}_{i \in I}$ be the T -components of U . Then for any $c \in C$, almost invariance gives that $U_i c = U_i$ for all but finitely many i . Since C is finite, it follows that U_i is T' -invariant for almost all i .

Suppose that one of these T' -invariant components U_i is not equal to $\emptyset$ or to T . Let $H = \{x \in T : U_i x = U_i\}$. Then H is an infinite subgroup of T . Choose $u \in U_i$ and $v \in T \setminus U_i$. Then $Hv \cap U_i = \emptyset$, and $Hu(u^{-1}v) \subseteq U(u^{-1}v)$, so $U(u^{-1}v)$ and U are not almost equal. Thus each of these T' -invariant components is either nontrivial or equal to $\emptyset$ or T . By changing U in the finitely many remaining components which are almost equal to $\emptyset$ or to T , we obtain a T -canonical set that is almost equal to U . $\square$

7.2 Weight

We now introduce the notion of *weight*, used heavily in the statement of Theorem 8.1 and more generally throughout the proof of Theorem 1.1. Broadly, the concept measures the level of interaction between the associated subgroups of a group pair $(G; \mathcal{S})$ and an arbitrary subgroup $T \leq G$.

Let $(G; S_1, \dots, S_m)$ be a group pair with a finite number of associated subgroups. For each $i = 1, \dots, m$, we have a restriction map

$$\text{res}_i : H^1(G; \mathbb{Z}_2 G) \rightarrow H^1(S_i; \mathbb{Z}_2 G)$$

induced by the inclusion $S_i \rightarrow G$. Denote by $N(G; S_1, \dots, S_m)$ the intersection $\bigcap_{j=1}^m \ker(\text{res}_j) \leq H^1(G; \mathbb{Z}_2 G)$.

Let T be a subgroup of G , and let $\{x_i \mid i \in I\}$ be a set of coset representatives for the right cosets in G/T . Then $\mathbb{Z}_2 G \cong \bigoplus_i \mathbb{Z}_2 T x_i$, and there is a natural embedding

$$j : \mathbb{Z}_2 G \rightarrow \prod_{i \in I} \mathbb{Z}_2 T x_i.$$

We consider the map

$$r : H^1(G; \mathbb{Z}_2 G) \xrightarrow{\text{res}} H^1(T; \mathbb{Z}_2 G) \xrightarrow{j^*} H^1(T; \prod_{i \in I} \mathbb{Z}_2 T x_i) \cong \prod_{i \in I} H^1(T; \mathbb{Z}_2 T x_i). \quad (7.1)$$

Definition 7.3. Let $(G; S_1, \dots, S_m)$ be a group pair such that $N(G; S_1, \dots, S_m) \neq \{0\}$, and let T be a subgroup of G . Then we define the **weight** $n(T) \in \mathbb{N} \cup \{\infty\}$ of T to be the minimal number of nontrivial components of $r(c)$ over all $c \neq 0$ in $N(G; S_1, \dots, S_m)$.

If further $r(N(G; S_1, \dots, S_m)) \neq \{0\}$, we define the **reduced weight** $\tilde{n}(T)$ of T to be the minimal number of nontrivial components of elements $d \neq 0$ in $r(N(G; S_1, \dots, S_m))$.

Note that the only difference between $n(T)$ and $\tilde{n}(T)$ is whether or not one considers elements $c \neq 0 \in N(G; S_1, \dots, S_m)$ with $r(c) = 0$, and thus $n(T) = \tilde{n}(T)$ exactly when $n(T) \neq 0$. Since all situations considered in this essay have $n(T) \neq 0$, we may use these two concepts interchangeably throughout this essay and simply write $n(T)$.

In a precise sense, the weight counts the number of right T -cosets on which a relevant almost invariant set restricts nontrivially, as the following proposition shows.

Proposition 7.1. *The map*

$$r : H^1(G; \mathbb{Z}_2 G) \rightarrow \prod_{i \in I} H^1(T; \mathbb{Z}_2 T x_i)$$

of Equation (7.1) has the following description in terms of almost invariant subsets:

$$r(\langle U \rangle) = \{\langle U \cap T x_i \rangle \mid i \in I\}.$$

Proof. Define $\overline{\mathbb{Z}_2 T x_i}$ analogously to $\overline{\mathbb{Z}_2 G}$ as the T -module $\text{Hom}(\mathbb{Z} T x_i, \mathbb{Z}_2)$, and define $\epsilon T x_i$ similarly. There is a natural map

$$\psi : \overline{\mathbb{Z}_2 G} \cong \prod_{i \in I} \overline{\mathbb{Z}_2 T x_i}$$

defined by $\psi(\chi_U) = (\chi_{U \cap T x_i})_{i \in I}$, which is clearly an isomorphism since the cosets $T x_i$ partition G . This descends to a morphism $\psi_\epsilon : \epsilon G \rightarrow \prod_{i \in I} \epsilon T x_i$, since every $\chi_{\{x\}} \in \iota(\mathbb{Z}_2 G)$ maps to a product of elements in $\iota_i(\mathbb{Z}_2 T x_i)$ for each i . Thus we get a commutative diagram

$$\begin{array}{ccccccc}
& & 0 & & 0 & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathbb{Z}_2 G & \longrightarrow & \overline{\mathbb{Z}_2 G} & \xrightarrow{\text{pr}} & \epsilon G \longrightarrow 0 \\
& & \downarrow j & & \downarrow \psi & & \downarrow \psi_\epsilon \\
0 & \longrightarrow & \prod_i \mathbb{Z}_2 T x_i & \longrightarrow & \prod_{i \in I} \overline{\mathbb{Z}_2 T x_i} & \xrightarrow{\{\text{pr}_i\}} & \prod_{i \in I} \epsilon T x_i \longrightarrow 0 \\
& & & & \downarrow & & \\
& & & & 0 & &
\end{array}$$

where j is the embedding from Section 7.2. We thus get a diagram of cohomology groups

$$\begin{array}{ccccccc}
(\overline{\mathbb{Z}_2 G})^G & \longrightarrow & (\epsilon G)^G & \xrightarrow{\delta} & H^1(G; \mathbb{Z}_2 G) & \longrightarrow & 0 \\
\downarrow \text{res} & & \downarrow \text{res} & & \downarrow \text{res} & & \\
(\overline{\mathbb{Z}_2 G})^T & \longrightarrow & (\epsilon G)^T & \longrightarrow & H^1(T; \mathbb{Z}_2 G) & \longrightarrow & 0 \\
\downarrow \psi_* & & \downarrow (\psi_\epsilon)_* & & \downarrow j_* & & \\
\prod_i (\overline{\mathbb{Z}_2 T x_i})^G & \longrightarrow & \prod_i (\epsilon T x_i)^G & \xrightarrow{\{\delta_i\}} & \prod_i H^1(T; \mathbb{Z}_2 T x_i) & \longrightarrow & 0
\end{array}$$

where the res maps are induced by the inclusion $T \rightarrow G$. Then we see from the definition of r and the diagrams above that

$$\begin{aligned}
r(\langle U \rangle) &= j_* \circ \text{res} \circ \delta(\text{pr}(\chi_U)) = \{\delta_i\} \circ (\psi_\epsilon)_* \circ \text{res}(\text{pr}(\chi_U)) \\
&= \{\delta_i\} \circ \{\text{pr}_i\} \circ \psi(\chi_U) = \{\delta_i \text{pr}_i\}(\chi_U \cap T x_i)_{i \in I} \\
&= \{\langle U \cap T x_i \rangle : i \in I\},
\end{aligned}$$

which completes the proof. $\square$

7.3 Stallings' Theorem and Cut Sets

The following celebrated theorem, proven first in [15] by Stallings in 1971, will be used several times in the proof of Theorem 1.1.

Theorem 7.1. *Let G be a finitely generated group. Then*

1. G has 0, 1, 2, or infinitely many ends.
2. G has 0 ends if and only if it is finite.
3. G has 2 ends if and only if it is virtually cyclic.
4. G has infinitely many ends if and only if it splits nontrivially over a finite subgroup.

There are a number of methods of proof of Stallings' theorem (in particular of (4) above, as the other cases are much easier). The general strategy of most proofs is to construct a family of subsets of G which mimic the structure of a tree using a condition similar to that given in Lemma 6.1, and then to build a tree upon which G has a natural action. Applying a Bass–Serre-type theorem similar to Theorem 6.1 gives a splitting of G ; this is the same general proof strategy employed in the proof of Theorem 8.1 in Section 8.

Many proofs of Stallings' theorem, including his original proof in the 1971 paper, tend to use the language of *cut sets* of Cayley graphs. This language will come up in the proof of Lemma 8.3, so some brief definitions and properties are given here.

Definition 7.4. Let G be a finitely generated group with finite generating set S , and let Γ denote the Cayley graph of G with respect to S . For a subset U of G , we define the **border** $\beta(U) = \{e \in E(\Gamma) \mid \iota(e) \in U, \tau(e) \in U^* \text{ or } \iota(e) \in U^*, \tau(e) \in U\}$ and the **breadth** $b(U) = |\beta(U)| \in \mathbb{N} \cup \{\infty\}$ of U (here U^* denotes the complement of U in G). U is called **connected** if the maximal subgraph of Γ generated by U is connected, and the maximal connected subsets of U are called the Γ -**components** of U .¹⁴

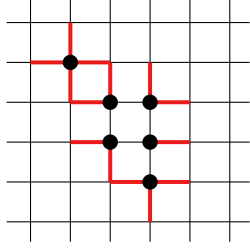


Figure 1: $G = \mathbb{Z}^2$

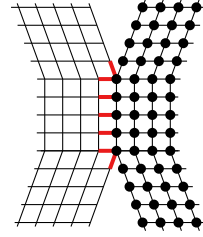


Figure 2: $\dim H^1(G; \mathbb{Z}_2 G) = \infty$

In both figures above, the black dots represent elements of $U \subset G$, and red edges represent elements of $\beta(U)$. Figure 2 illustrates how the existence of a large set U with finite $\beta(U)$ can detect a splitting of G .

Finite breadth is the graph-theoretic analogue of almost invariance:

Proposition 7.2. Let U be a subset of a finitely generated group G . Then $b(U) < \infty$ if and only if U is almost G -invariant.

Proof. An edge beginning in U and ending in U^* (i.e. an element of $\beta(U)$) exactly corresponds to an element $u \in U$ such that $us \notin U$ for some $s \in S$; if there are only finitely many elements of $\beta(U)$, then there are only finitely many such elements in U , and vice versa. $\square$

The following are lemmas concerning almost invariant sets that will be useful in some proofs in Section 8:

Lemma 7.2. Let A, B be sets of vertices of a graph Γ . If B is connected and $B \cap \beta(A) = \emptyset$, then either $B \subseteq A$ or $B \subseteq A^*$.

Proof. If not, then we can choose $v \in B \cap A$ and $w \in B \cap A^*$. Since B is connected, there exists a path from v to w in B . This path crosses from A to A^* at some point, and so $B \cap \beta(A) \neq \emptyset$. $\square$

Lemma 7.3. Let A, B be sets of vertices of a connected graph Γ . If there exist connected sets of vertices C, D of Γ with $C \cap D = \emptyset$, $\beta(A) \subseteq C$, and $\beta(B) \subseteq D$, then one of the intersections (a) $A \cap B$, (b) $A \cap B^*$, (c) $A^* \cap B$, or (d) $A^* \cap B^*$ is empty.

Proof. Since $D \cap \beta(A) = C \cap \beta(B) = \emptyset$, either $D \subseteq A$ or $D \subseteq A^*$ and either $C \subseteq B$ or $C \subseteq B^*$ by Lemma 7.2. By symmetry, we may suppose that $D \subseteq A^*$ and $C \subseteq B^*$. Now, $(A \cap B)^* \neq \emptyset$, so to see that $A \cap B = \emptyset$ it is enough to show that $\beta(A \cap B) = \emptyset$. Any edge e in $\beta(A \cap B)$ has one vertex in $A \cap B$, and the other vertex in one of the sets (b)-(d). In case (b), both vertices of e are in B , which is impossible since $\beta(A) \subseteq C \subseteq B^*$. Similarly, case (c) is impossible since $\beta(B) \subseteq D \subseteq A^*$. Case (d) would give an edge in $\beta(A) \cap \beta(B) \subseteq C \cap D$, which contradicts $C \cap D = \emptyset$. $\square$

Lemma 7.4 (Lemma 2.8 in [6]). Let E_0 and E_1 be almost invariant subsets of the finitely generated group G (with Cayley graph Γ). Then for almost all $g \in E_0$, either $E_1 g \subseteq E_0$ or $E_1^* g \subseteq E_0$.

¹⁴Note that all of these concepts are defined for any graph Γ . However, the only cases that will arise in this paper will be those concerning the Cayley graph of a group.

Proof. Let C_0 and C_1 be finite connected subsets of Γ such that $\beta(E_i) \subseteq C_i$ for $i = 0, 1$. Since E_0 is almost invariant, we have that $cg \in E_0$ for almost all $g \in E_0$ and each $c \in C_1$. Thus $C_1g \subseteq E_0$ for almost all $g \in E_0$. Since C_0 is finite, we must also have that $C_0 \cap C_1g = \emptyset$ for almost all $g \in G$. If g is such that $C_0 \cap C_1g = \emptyset$, then applying Lemma 7.3 (with $A = E_0$, $B = E_1g$) gives that one of $E_0 \cap E_1g$, $E_0 \cap E_1^*g$, $E_0^* \cap E_1g$, $E_0^* \cap E_1^*g$ is empty. If additionally $C_1g \subseteq E_0$, the first case is immediately ruled out, and $E_0 \cap E_1^*g = \emptyset$ would imply that $E_0 \subseteq E_1g$ (so that $E_0 = E_1g$). Thus we have that either $E_0^* \cap E_1g = \emptyset$ or $E_0^* \cap E_1^*g = \emptyset$. The first case gives that $E_1g \subseteq E_0$, and the second gives that $E_1^*g \subseteq E_0$. $\square$

Lemma 7.5 (Lemma 2.9 in [6]). *Let G be a finitely generated group. If E is a nontrivial almost invariant subset of G such that $\{g \in G \mid Eg \stackrel{a}{=} E\}$ is infinite, then G has two ends.*

Proof. Since E and E' are infinite, one of $\{g \in E \mid Eg \stackrel{a}{=} E\}$ or $\{g \in E^* \mid Eg \stackrel{a}{=} E\}$ is infinite, and so we may assume without loss of generality that $\{g \in E \mid Eg \stackrel{a}{=} E\}$ is infinite, and by changing E if necessary we may assume that the identity element e is in E . By Lemma 7.4 (with $E_0 = E \setminus \{e\}$ and $E_1 = E$), for almost all $g \in E$ either $Eg \subseteq E \setminus \{e\}$ or $E^*g \subseteq E \setminus \{e\}$, so we can choose some c satisfying one of these conditions. Since $Ec \stackrel{a}{=} E$, we must have the case $Ec \subseteq E \setminus \{e\}$.

For $n > 0$, $Ec^n \subseteq Ec \subset E$, so $c^n \neq e$. Since $e \in E$ and $Ec^n \subset E$, $c^n \in E$ for all $n > 0$, and since $Ec^n \subseteq E \setminus \{e\}$ we must have that $c^{-n} \in E^*$ for $n > 0$. We know that $\bigcap_{n>0} Ec^n \neq \emptyset$, since if $d \in \bigcap_{n>0} Ec^n$ then $c^{-n} \in Ed^{-1} \stackrel{a}{=} E$ for all $n > 0$, which contradicts the fact that $c^{-n} \in E^*$ for $n > 0$. By a similar argument, $\bigcap_{n>0} E^*c^{-n} = \emptyset$. Thus $E = \bigcup_{n \geq 0} (Ec^n \setminus Ec^{n+1}) = \bigcup_{n \geq 0} (E - Ec)c^n$, and similarly $E^* = \bigcup_{n \geq 0} (E^* \setminus E^*c^{-1})c^{-n}$. Since $E \stackrel{a}{=} Ec$, both $E \setminus Ec$ and $E^* \setminus E^*c^{-1}$ are finite. Any representative of any coset of $\langle c \rangle$ in G is either in E or E^* , and thus there exists some representative for the same coset in $E \setminus Ec$ or $E^* \setminus E^*c^{-1}$. Thus $\langle c \rangle$ is of finite index in G , so G has two ends by Stallings' theorem. $\square$

8 Simultaneous Splitting

This section will be an adaptation of the proof of the main theorem of [13] to the specific settings needed in [10]. In particular, rather than proving that theorem in full generality, this section will tackle directly the two cases used in [10] (Corollary 3.2. and Corollary 3.6. in [13]), and will make use of the many assumptions made about the group pairs and associated subgroups considered in the applications of these corollaries. We begin with a definition.

Definition 8.1. *Let (G, S) be a group pair, and suppose $G = G_1 *_K G_2$ or $G = G_1 *_K p$ is a splitting of G . The splitting is said to be **adapted** to $(G, \mathcal{S})$ if each $S \in \mathcal{S}$ is conjugate into a vertex group of the splitting.*

Then the modified theorem then goes as follows:

Theorem 8.1. *Let G be a finitely generated group, and $(G; \mathcal{S}) = (G; S_1, \dots, S_m)$ a finite group pair. Suppose that a torsion-free subgroup T of G is such that $r(N(G; S_1, \dots, S_m)) \neq 0$ and $n = \tilde{n}(T) > 0$. Then we have the following:*

(A) *If $n = 1$, G is torsion-free, and T is a free group of rank $\ell > 1$, then there exist splittings of G and T , such that the splitting of G is adapted to the pair $(G; \mathcal{S})$, of one of the following forms (In the below, H_i and L_i are subgroups of H and L respectively):*

1. $G = H * L$ and $T = H_1 * L_1$.
2. $G = H * \langle p \rangle$ and $T = H_1 * p H_2 p^{-1}$.

(B) If G is a free group of rank $\ell > 1$, $n = 2$, and T is a cyclic group, then there exists a splitting of G , adapted to the pair $(G; \mathcal{S})$, of one of the following forms:

1. $G = G_1 * G_2$ and $T = \langle g_1 g_2 \rangle$, where $e \neq g_i \in G_i$ for $i = 1, 2$.
2. $G = G_1 * \langle p \rangle$ and $T = \langle p g_1 p^{-1} g_2 \rangle$, where $e \neq g_1, g_2 \in G_1$.

8.1 Proof of Theorem 8.1

Throughout the proof, any assertion about G or T made before the cases split should be understood as following from the hypotheses common to both cases (A) and (B).

First note that since $r : H^1(G; \mathbb{Z}_2 G) \rightarrow \prod_i H^1(T; \mathbb{Z}_2 T x_i)$ is a nonzero map, we must have that $H^1(G; \mathbb{Z}_2 G) \neq 0$. Since $H^1(G; \mathbb{Z}_2 G)$ is nontrivial, by Stallings' Theorem (7.1), G has either 2 or infinitely many ends, and G has 2 ends if and only if G is virtually cyclic. In case (A), G contains a free subgroup of rank $\ell > 1$ and is not virtually cyclic. In case (B), G is a free group of rank $\ell > 1$ and similarly is not virtually cyclic. We therefore know that G has infinitely many ends.

The following lemma will be used in order to make sure that the resulting splittings are adapted to the pair $(G; \mathcal{S})$. This will be done by embedding each S_i into a larger group with trivial first cohomology. We proceed as follows:

Lemma 8.1. *Let G be a group with infinitely many ends satisfying the conditions of Theorem 8.1. Then G embeds into a finitely generated group A with the following properties:*

1. $\text{res} : H^1(A; \mathbb{Z}_2 A) \rightarrow H^1(G; \mathbb{Z}_2 A)$ is injective.
2. $\text{im}(\text{res}) \cap \{c \in N(G; S_1, \dots, S_m) \mid r(c) \text{ has } \tilde{n}(T) \text{ components}\} \neq \emptyset$.¹⁵
3. For each S_j for $j = 1, \dots, m$, there exists a finitely generated subgroup R_j of A with one end such that $S_j \subseteq R_j \subseteq A$.

Proof. Since G is finitely generated, we can certainly embed each S_j into a finitely generated group R_j such that R_j has one end (for instance, set $R_j = G \times \mathbb{Z} \times \mathbb{Z}$). Set $G_0 = G$, and inductively define $G_i = G_{i-1} *_{S_i} R_i$ for $1 \leq i \leq m$. The Mayer-Vietoris sequence for amalgamated free products (obtained by, applying Mayer-Vietoris for spaces to the associated graph of spaces of this decomposition) yields the exact sequences

$$0 \rightarrow H^1(G_i; \mathbb{Z}_2 G_m) \xrightarrow{(\text{res}, \text{res})} H^1(G_{i-1}; \mathbb{Z}_2 G_m) \oplus H^1(R_i; \mathbb{Z}_2 G_m) \xrightarrow{\text{res} \oplus \text{res}} H^1(S_i; \mathbb{Z}_2 G_m) \rightarrow \dots$$

Since the R_i are one-ended, $H^1(R_i; \mathbb{Z}_2 G_m) = 0$ for all i , so in fact we get exact sequences

$$0 \rightarrow H^1(G_i; \mathbb{Z}_2 G_m) \xrightarrow{\text{res}_i} H^1(G_{i-1}; \mathbb{Z}_2 G_m) \xrightarrow{\text{res}} H^1(S_i; \mathbb{Z}_2 G_m) \rightarrow \dots$$

Indeed, then, since each of the restriction maps from $H^1(G_i, \mathbb{Z}_2 G_m)$ to $H^1(G_{i-1}, \mathbb{Z}_2 G_m)$ are injective, we conclude inductively that each map $\text{res}_i : H^1(G_i; \mathbb{Z}_2 G_m) \rightarrow H^1(G; \mathbb{Z}_2 G_m)$ is injective. Recalling the definition of $N(G; S_1, \dots, S_i)$, we see that exactness at the middle term of the above sequence also gives that $\text{im}(\text{res}_i) \cap H^1(G; \mathbb{Z}_2 G) = N(G; S_1, \dots, S_i)$. $\square$

¹⁵Here consider $H^1(G; \mathbb{Z}_2 G)$ as a submodule of $H^1(G; \mathbb{Z}_2 A)$, so that this intersection makes sense.

We can thus let A be as in the result of Lemma 8.1. Let $c \in H^1(A; \mathbb{Z}_2 A)$ be such that $d = \text{res}(c)$ is an element of $N(G; \mathcal{S}) \subseteq H^1(G; \mathbb{Z}_2 G) \subseteq H^1(G; \mathbb{Z}_2 A)$ with $n(T)$ components (considering again $H^1(G; \mathbb{Z}_2 G)$ as a submodule of $H^1(G; \mathbb{Z}_2 A)$).

Recalling the characterization of elements of $H^1(A; \mathbb{Z}_2 A)$ in terms of almost invariant subsets, choose $E_A \subset A$ with $\langle E_A \rangle = c$, where E_A is almost A -invariant. Since T is not locally finite and E_A is certainly almost T -invariant, we see that E_A is almost T -canonical by Lemma 7.1. We can then change the representative E_A of $c = \langle E_A \rangle$ if necessary so that E_A is T -canonical. Then, again recalling the characterization of $\text{res} : H^1(A; \mathbb{Z}_2 A) \rightarrow H^1(G; \mathbb{Z}_2 A)$ in terms of almost invariant sets, we have that $d = \langle E \rangle$, where $E = E_A \cap G$.

Lemma 8.2. *The following statements are true:*

1. $E_A = E \cup \bigcup_{i \in I} G y_i$ for some indexing set I and some $y_i \in A \setminus G$.
2. $E = \left(\bigcup_{0 \leq i < n} E_i x_i \right) \cup \left(\bigcup_{j \in J} T x_j \right)$ for some indexing set J , some $x_0, \dots, x_{i-1}$, and some choices of $x_j \in G \setminus \bigcup_{0 \leq i < n} T x_i$ for each $j \in J$.
3. $E \subset G$ and $E_i \subset T$ are not trivial for $0 \leq i < n$.
4. E_A is almost A -invariant and T -canonical.

Conversely, if a set E_A (and $E = E_A \cap G$) satisfy (1)-(4), then $\langle E_A \rangle \in H^1(A; \mathbb{Z}_2 A)$ is such that $\text{res}(\langle E_A \rangle)$ is an element of $N(G; \mathcal{S})$ with $n(T)$ components.

Proof. (4) is true from the definitions. For (1), first decompose A as $A = \bigsqcup_k G y_k$. note that the fact that $\text{res}(c)$ lies in $H^1(G; \mathbb{Z}_2 G)$ means that the composition

$$H^1(A; \mathbb{Z}_2 A) \xrightarrow{\text{res}} H^1(G; \mathbb{Z}_2 A) \xrightarrow{j^*} \prod_k H^1(G; \mathbb{Z}_2 G y_k)$$

has image lying entirely in $H^1(G; \mathbb{Z}_2 G)$. Since E is T -canonical, it is also G -canonical, so for each k we have that $E_A \cap G y_k$ is either equal to $\emptyset$ or to $G y_k$. Letting I be the indexing set of the k such that this set is equal to $G y_k$ gives the needed form.

To see (2), note that since $r(d)$ has exactly n nontrivial components, there are exactly n cosets $T x_i$ for which the component of E in $T x_i$ is nontrivial (following the characterization of r in [something](#)). The remaining components are trivial, which since E is T -canonical means that they are equal to $\emptyset$ or to $T x_i$; the latter case gives the $T x_j$. (3) is clear from the definitions of E and E_i in part (2).

Conversely, an E_A satisfying (1)-(4) induces a T -canonical element $\langle E_A \rangle = c \in H^1(A; \mathbb{Z}_2 A)$ by (4). By (1), $\text{res}(c) \in H^1(G; \mathbb{Z}_2 G)$. We know therefore that $\text{res}(c) \in N(G; S_1, \dots, S_m)$; indeed, at least for elements of $H^1(A; \mathbb{Z}_2 A)$ the diagram

$$\begin{array}{ccc} H^1(A; \mathbb{Z}_2 A) & \xrightarrow{\text{res}} & H^1(G; \mathbb{Z}_2 A) \\ \text{res}_i \downarrow & & \downarrow \text{res}_i \\ H^1(R_i; \mathbb{Z}_2 A) & \xrightarrow{\text{res}} & H^1(S_i; \mathbb{Z}_2 A) \end{array}$$

commutes, and $H^1(R_i; \mathbb{Z}_2 A) = 0$ for all i , so indeed c must be in the kernel of each of the maps res_i (even when restricted to $H^1(G; \mathbb{Z}_2 G)$). The fact that $r(d)$ has n nontrivial components is exactly the statement of (2) in the language of almost invariant sets. $\square$

Definition 8.2. Let $(G; \mathcal{S})$ and T be as in Theorem 8.1, and let A be as in the result of Lemma 8.1. If $S_A \subseteq A$ is an almost invariant subset such that S_A (and $S = S_A \cap G$) satisfy conditions (1)-(4) of Lemma 8.2, we will write in the following that S_A has **form $\star$** .

Note that any set S that is almost of form $\star$ (in the sense that $S \stackrel{a}{=} E$ for a set E of form $\star$) also satisfies conditions (1)-(4) with the equalities in (1) and (2) replaced by almost equalities.

Lemma 8.3. Let $(G; \mathcal{S})$ and T be as in Theorem 8.1, and let A be as in the result of Lemma 8.1. Then there exists a set S_A , almost of form $\star$, such that for all $x \in A$ at least one of $S_A \cap S_Ax$, $S_A \cap S_A^*x$, $S_A^* \cap S_Ax$, and $S_A^* \cap S_A^*x$ is empty (where S_A^* denotes the complement of S_A in A).

Proof. Choose a finite generating set C of A containing a set of generators for T . Let Γ be the Cayley graph of A with respect to C . Choose a set S_A of form $\star$ with minimal breadth $b = b(S_A)$ (such a set can be chosen since $H^1(A; \mathbb{Z}_2A) \neq 0$, and b is finite by Proposition 7.2). Assume that for some $x \in A$, the intersections (a) $S_A \cap S_Ax$, (b) $S_A \cap S_A^*x$, (c) $S_A^* \cap S_Ax$, and (d) $S_A^* \cap S_A^*x$ are all infinite.

Suppose for contradiction that x is not in G . Using Lemma 8.2 (1), write $S_A = S \cup \bigcup_{i \in I} Gy_i$. Then the only nontrivial G -coset of S_Ax is $Sx \subseteq Gx$. Then we have four cases, depending on which of Gx and Gx^{-1} appear in $\{Gy_i\}_{i \in I}$. If $Gx, Gx^{-1} \notin \{Gy_i\}_{i \in I}$, then $S_A \cap S_Ax$ is a union of G -cosets, as $E \cap \bigcup_i Gy_i x = \bigcup_i Gy_i \cap Ex = \emptyset$. This means that $\text{res}(\langle S_A \cap S_Ax \rangle) = 0$ in $H^1(G; \mathbb{Z}_2A)$, which is impossible since $\text{res} : H^1(A; \mathbb{Z}_2A) \rightarrow H^1(G; \mathbb{Z}_2A)$ is injective (Lemma 8.2 (1)). A similar argument for the other three possibilities shows that this situation is not possible in general.

Therefore x is in G , and so $Sx, S^*x \subseteq G$ and we can thus write each of the four intersections as a union of their corresponding intersection in G and G -cosets: e.g. $S_A \cap S_Ax = (S \cap Sx) \cup \bigcup_j G_j y_j$ for some $y_j \in A$, and similarly for (b)-(d). Let (a') = $S \cap Sx$ denote the G -component of (a), and similarly define (b')-(d'). Recall the decomposition $S = \bigcup_{i=0}^{n-1} S_i x_i \cup \bigcup_j T z_j$ in Lemma 8.2 (2), and also note that $S^* = \bigcup_{i=0}^{n-1} S_i^* x_i \cup \bigcup_k T z_k$ for some other cosets $k \in K$, where each S_i^* is also nontrivial. Indeed, then, if the T -component of some coset of one of (a')-(d') is nontrivial, it must be the case that this coset is of the form $S_i x_i$, $S_i^* x_i$, $S_i x_i x$, or $S_i^* x_i x$ for some $0 \leq i < n$. We thus conclude that, in total, the sets (a')-(d') contain at most $4n$ nontrivial T -components.

Now, since S_A (and thus S_A^*) are both almost A -invariant, we know that $S_A \cap S_Ax$ is almost A -invariant, and in fact (since $x \in G$) that $\text{res}(\langle S_A \cap S_Ax \rangle) = \langle S \cap Sx \rangle \in H^1(G; \mathbb{Z}_2G)$, and therefore $\langle S \cap Sx \rangle \in N(G; S_1, \dots, S_m)$ (by the argument at the end of the proof of Lemma 8.2 10.2). By the minimality of n , $S \cap Sx$ must have at least n nontrivial T -components. Finally, the decomposition of $S \cap Sx$ shows that it is T -canonical, and thus that $S_A \cap S_Ax$ is also T -canonical. The same argument applies to (b)-(d): combining this with the upper bound in the previous paragraph tells us that each of the four intersections contain *exactly* n nontrivial T -components. Lemma 8.2 10.2 thus tells us that all of (a)-(d) are of form $\star$.

By minimality, we know that $b(S_A \cap S_Ax) \geq b$, and similarly for (b)-(d). Every element η in $\beta(S_A) \cup \beta(S_Ax)$ must occur in exactly two of the four sets

$$\beta(S_A \cap S_Ax), \quad \beta(S_A \cap S_A^*x), \quad \beta(S_A^* \cap S_Ax), \quad \beta(S_A^* \cap S_A^*x),$$

depending on whether $\eta \in \beta(S_A) \cap \beta(S_Ax)$ (so that $\eta \in \beta(S_A \cap S_Ax) \cap \beta(S_A^* \cap S_A^*x)$) or $\eta \in \beta(S_A) \triangle \beta(S_Ax)$ (so that $\eta \in \beta(S_A \cap S_A^*x) \cap \beta(S_A^* \cap S_Ax)$). Then

$$b(S_A \cap S_Ax) + \dots + b(S_A^* \cap S_A^*x) = 2|\beta(S_A) \cup \beta(S_Ax)| \leq 2b(S_A) + 2b(S_Ax) = 4b,$$

so indeed we must have that $b(S_A \cap S_A x) = b$, and similarly for (b)-(d).

Fix once and for all some element $x_0 \in A$, and choose the set out of (a)-(d) that contains x_0 ; denote it S'_A . If there exists an $x' \in A$ such that all four intersections $S'_A \cap S'_A x'$, $S'_A \cap S'^*_A x'$, $S'^*_A \cap S'_A x'$, $S'^*_A \cap S'^*_A x'$ are infinite, repeat this process. If this process did not stop, we could construct an infinite, strictly descending sequence

$$S_A \supset S'_A \supset \dots \supset S_A^{(k)} \supset \dots$$

of sets of form $\star$ and breadth b containing x_0 . Let $\bar{S}_A = \bigcap_{k \in \mathbb{N}} S_A^{(k)}$.

Choose $e_0 \in \beta(\bar{S}_A)$. Then eventually one vertex of e_0 is in $S_A^{(k_0)}$ and one is not (i.e. $e_0 \in \beta(S_A^{(k_0)})$), and this stays true for all $\ell \geq k_0$ since the sequence is decreasing. Thus we can choose $e_1 \in \beta(\bar{S}_A)$ and find k_1 such that $e_0, e_1 \in \beta(S_A^{(k_1)})$. Continuing this way, we see that for any *finite* subset E of $\beta(\bar{S}_A)$, $E \subseteq \beta(S_A^{(k)})$ for some k . Since $|\beta(S_A^{(k)})| = b$ for *all* k , we conclude that $\beta(\bar{S}_A)$ is finite and that $\beta(\bar{S}_A) \subseteq \beta(S_A^{(n_0)})$ for some n_0 .

We claim that $\bar{S}_A$ is infinite. Indeed, if $\bar{S}_A$ contained an element of a nonempty trivial component Ty of $S_A^{(n_0)}$, then it contains all of Ty since C contains generators for Ty and thus $Ty \subseteq S_A^{(n_0)}$ is connected. However, if $\bar{S}_A$ were finite and contained only in the nontrivial T -components of $S_A^{(n_0)}$, then $S_A^{(n_0)} \setminus \bar{S}_A$ would still be of form $\star$, and since $\bar{S}_A$ is nonempty would have strictly smaller breadth, contradicting the minimality of b .

Thus $\bar{S}_A$ is infinite, and is almost A -invariant since it has finite breadth. Let

$$\Omega = \{Ty \mid \bar{S}_A \cap Ty \neq \emptyset \text{ and } \bar{S}_A \cap Ty \neq Ty\}$$

be the set of non-trivial T -components of $\bar{S}_A$. Then for any finite subset $\{Ty_i\}_{i=1}^\ell \subseteq \Omega$, there exist sets $\{S_A^{(N_i)}\}_{i=1}^\ell$ such that $S_A^{(N_i)}$ has a nontrivial Ty_i component for each i . Then $S_A^{(N)}$ has a nontrivial Ty_i component for all $1 \leq i \leq \ell$, where $N = \max\{N_i\}_{i=1}^\ell$; this shows that $|\Omega| \leq n$. However, since $\text{res}(\langle \bar{S}_A \rangle) \in H^1(G; \mathbb{Z}_2 G)$ and $\langle \bar{S}_A \rangle \neq 0$, we know that $\bar{S}_A$ must have at least n nontrivial T -components. Hence $|\Omega| = n$. Since each $S_A^{(i)}$ is T -canonical, $\bar{S}_A$ is as well. Thus $\bar{S}_A$ has form $\star$, so by the minimality of b , $b(\bar{S}_A) \geq b$. Since $\beta(\bar{S}_A) \subseteq \beta(S_A^{(n_0)})$ and both sets are finite, this gives that $\beta(\bar{S}_A) = \beta(S_A^{(n_0)})$. Since $\bar{S}_A \subseteq S_A^{(n_0)}$, we must in fact have that $\bar{S}_A = S_A^{(n_0)}$. To see this, note that if v is any element in $\beta(S_A^{(n_0)} \setminus \bar{S}_A)$, then either v is adjacent via an edge e_1 to a vertex in $\bar{S}_A$ (so that $e_1 \in \beta(\bar{S}_A) \setminus \beta(S_A^{(n_0)})$), or v is adjacent via an edge e_2 to a vertex in $S_A^{(n_0)*}$ (so that $e_2 \in \beta(S_A^{(n_0)}) \setminus \beta(\bar{S}_A)$). Neither case is possible, so $\beta(S_A^{(n_0)} \setminus \bar{S}_A) = \emptyset$, which is only possible if $S_A^{n_0} = \bar{S}_A$.

This is a contradiction since the sequence $\{S_A^{(k)}\}$ is strictly decreasing. Therefore the process described earlier halts after finitely many steps, which exactly gives a set S_A of form $\star$ such that one of the four intersections is finite for all $x \in A$.

Consider the set $S'_A = \{x \in A \mid S_A x \stackrel{a}{\subseteq} S_A \text{ or } S_A^* x \stackrel{a}{\subseteq} S_A\}$. This set satisfies the necessary condition on intersections: for instance, if $S_A^* \cap S_A x$ is finite, then $S_A x \stackrel{a}{\subseteq} S_A$, and so either $S_A^* y x \stackrel{a}{\subseteq} S_A$ or $S_A y x \stackrel{a}{\subseteq} S_A$ for all $y \in S'_A$, which means that $S'_A x \subseteq S'_A$ and thus $S'_A * \cap E'_A x = \emptyset$.

Finally, we want to show that $S'_A \stackrel{a}{=} S_A$, which will complete the proof. Applying Lemma 7.4 with

$E_0 = E_1 = S_A$ gives that $S_A x \subseteq S_A$ or $S_A^* x \subseteq S_A$ for almost all $x \in S_A$, which shows that $S_A \setminus S'_A$ is finite. Applying Lemma 7.4 again with $E_0 = E_1 = S_A$, we get that for almost all $x \in S_A^*$, one of $S_A^* x \subseteq S_A^*$ or $S_A x \subseteq S_A^*$ holds. If such an x also satisfied the defining condition for S'_A , we would either have that $S_A^* x \subseteq S_A^*$ and $S_A x \subseteq S_A$, which would imply that $S_A x \stackrel{a}{=} S_A$, or we would have that $S_A x \subseteq S_A^*$ and $S_A^* x \subseteq S_A$, which would also imply that $S_A x \stackrel{a}{=} S_A$. If there were infinitely many such $x \in S'_A \cap S_A^*$, then there would be infinitely many x such that $S_A x \stackrel{a}{=} S_A$, which by Lemma 7.5 would mean that G is two-ended. Since G has infinitely many ends, this cannot occur, so $S'_A \cap S_A^*$ is finite. Therefore $S'_A \stackrel{a}{=} S_A$. Replacing S_A by S'_A , we obtain the desired set. $\square$

8.2 Construction of Trees

Let E_A be a set satisfying Lemma 8.3, with $E = E_A \cap G$ and $E_j, 0 \leq j < n$, as in Lemma 8.2. Define

$$\begin{aligned}\epsilon_A &= \{E_A x, E_A^* x \mid x \in A\}, \\ \epsilon_G &= \{E x, E^* x \mid x \in G\}, \\ \epsilon_T &= \{E_j x, E_j^* x \mid x \in T, 0 \leq j < n\}.\end{aligned}$$

Proposition 8.1. *There exists a tree X_A with $E(X_A) = \epsilon_A$.*

Proof. This will follow from Lemma 6.1, provided we show that $(\epsilon_A, \subseteq, *)$ satisfies the conditions (1)-(4) of the latter part of the lemma. Conditions (1) and (4) are trivial. Condition (3) is guaranteed from Lemma 8.3. (2) follows from the argument of Lemma 4.2 in [8], which proceeds as follows:

For any two sets $E, E' \in \epsilon_A$, A is the union of $E \cap E'$, $E^* \cap E'$, $E \cap E'^*$, and $E^* \cap E'^*$, so by applying Lemma 7.4, we see that $E_A g$ or $E_A^* g$ is contained in one of these four sets for almost all $g \in G$. Suppose $E \subseteq E'$. Then if g is such that $E \subset E_A g \subset E'$, with $E_A g \neq E, E'$, then we know that neither $E_A g$ nor $E_A^* g$ is contained in $E \cap E'$, $E^* \cap E'$, $E \cap E'^*$, or $E^* \cap E'^*$. Thus there are only finitely many such g , so condition (2) is satisfied. $\square$

Note that there is an obvious action of A on X_A by right multiplication.

Lemma 8.4. *If $E x \subset E^*$ for some $x \in G$, then $E_A x \subset E_A^*$.*

Additionally, recalling the decomposition of E into $E_0, \dots, E_{n-1}$ and T -cosets in Lemma 8.2, we see that if $E_p t \subset E_q$ for some $0 \leq p, q < n$ and some $t \in T$, then $E x_p^{-1} t \subset E^ x_q^{-1}$.*

Proof. For all $x \in G$, one of $E_A x \subseteq E_A$, $E_A^* x \subseteq E_A$, $E_A x \subseteq E_A^*$, $E_A^* x \subseteq E_A^*$ holds, by Lemma 8.3. If $E x \subset E^*$, then only $E_A x \subset E_A^*$ is possible. The same proof works for the second statement. $\square$

Proposition 8.2. *There exist trees X_G, X_T with $E(X_G) = \epsilon_G$ and $E(X_T) = \epsilon_T$.*

Proof. As in the previous proposition, we verify the four conditions of Lemma 6.1. Conditions (1) and (4) are again trivial. In both cases, condition (3) follows immediately from Lemma 8.3 and the fact that G and T are subsets of A . Condition (2) follows from the fact that ϵ_A satisfies the same condition, as any chain of inclusions in ϵ_G lifts to a chain of inclusions in ϵ_A via Lemma 8.4, and similarly for ϵ_T . $\square$

We similarly get actions of G and T on X_G and X_T respectively.

Lemma 8.5. *Define subgroups $K = \{x \in G \mid E x = E\}$ and $K' = \{x \in G \mid E x = E \text{ or } E^*\}$ of G . Then K and K' are both trivial, and thus the action of G on X_G is without edge inversions.*

Proof. Since $K \subseteq E$, E is a union of left K -cosets, as is xE for any $x \in G$. If K were infinite, then the symmetric difference $E \Delta xE$ would be a union of left K -cosets, and can thus only be finite if it is empty. However, recall that $E = E_A \cap G$ is almost G -invariant, so that $xE \stackrel{a}{=} E$ for all $x \in G$. These facts together would imply that $xE = E$ for all $x \in G$, so that E is trivial, a contradiction. Thus K is finite. The subgroup K has index at most two in K' .

In both case (A) and case (B), G is torsion-free, so K and K' must both be trivial. $\square$

G then acts without edge inversions on X_G . We may then consider the quotient graph $Y_G = X_G/G$, and we can thus apply the fundamental theorem of Bass–Serre theory (Theorem 6.1) to obtain a decomposition of G as a graph of groups. Indeed, by the construction of ϵ_G , the underlying graph of this graph of groups will have one edge, corresponding to the G -orbit of E .

Define the following, in accordance with the construction of the tree X_G in Lemma 6.1:

$$\begin{aligned} H &= \text{stab}_G(\iota(E)) = \{g \in G \mid Eg = E \text{ or } E^*g \subseteq E\} \\ L &= \text{stab}_G(\tau(E)) = \{g \in G \mid Eg = E \text{ or } Eg \subseteq E^*\} \end{aligned}$$

Let p be the stable letter corresponding to the unique edge of Y_G . In total, we get the following decomposition:

Proposition 8.3. *The group G splits as follows:*

- If $\iota(E) \neq \tau(E)$, then $G = H *_K L$.
- If $\iota(E) = \tau(E)$, then $G = H *_K p$ for the stable letter p . In this case, then, $L = pHp^{-1}$.

Since E is not trivial, this decomposition of G is not trivial.

Let us note the precise circumstances under which Y_G may have only one vertex. The vertices of X_G are the $(\sim)$ -equivalence classes of ϵ_G . The action of G on X_G can certainly take the class $[E]$ to any other class $[Et]$ by multiplication, and similarly for $[E^*]$ and $[E^*t]$. The question then reduces to whether there exists a t with $[Et] = [E^*]$, such that $Et \sim E^*$. This would mean that either $Et = E^*$ or that $Et \subseteq E$ with $Et \neq E$. If this is the case, then the graph of groups Y_G is an HNN extension, and we can choose any such t to be the stable letter.

Now we split the cases (A) and (B).

Proof of Case (A). In this case, the weight is 1, so there is exactly one nontrivial T -component E_0 of E . As for G , let $K = \{t \in T : E_0t = E_0\}$ and $K' = \{t \in T : E_0t = E_0 \text{ or } E_0^*\}$. Then K and K' are trivial by the same argument as above, since T is torsion-free.

Thus T acts on the tree X_T without edge inversions. Let Y_T denote the quotient X_T/T . By the construction of ϵ_T , this graph will have only one edge pair, (E_0, E_0^*) . We apply Theorem 6.1 again to obtain a decomposition of T as a graph of groups.

Let h be the stable letter corresponding to the edge E_0 in Y_T , recalling that $h = 1$ if Y_T is a tree. Recalling the construction of the tree in the proof of Lemma 6.1, we see that

$$\begin{aligned} T_{V_1} &= \text{stab}_T(\iota(E_0)) = \{t \in T \mid E_0t = E_0 \text{ or } E_0^*t \subseteq E_0\} \\ hT_{V_2}h^{-1} &= \text{stab}_T(\tau(E_0)) = \{t \in T \mid E_0t = E_0 \text{ or } E_0t \subseteq E_0^*\} \end{aligned}$$

If G splits as $G = H * L$ as above, then it is clear from the definitions of T_{V_1} and H that $T_{V_1} \subseteq H$, and similarly that $hT_{V_2}h^{-1} \subseteq L$. Now suppose there existed a nontrivial stable letter h such that $E_0h \subset E_0$. Lemma 8.4, we would have $Ex_0^{-1}h \subset E^*x_0^{-1}$, so that $x_0^{-1}hx_0$ is a stable letter for G . Thus the splitting cannot have a stable letter, so $h = 1$ and T splits as $T = T_{V_1} * T_{V_2}$ with $T_{V_1} \subseteq H$ and $T_{V_2} \subseteq L$.

If G splits as $G = H *_{1,p}$, then as before we have that $T_{V_1} \subseteq H$. Consider the stable letter p satisfying $Ep \subset E$. Since there are only finitely many infinite T -cosets of E , and multiplication by p cannot make a trivial T -coset nontrivial, p must send the coset containing E_0 to itself, so that $E_0x_0p \subset E_0x_0$. Then $E_0x_0px_0^{-1} \subset E_0 \subset T$, so we can replace p by $x_0px_0^{-1}$ so that $p \in T$. Thus we know that T splits as $T_{V_1} *_{1,p}$, with $T_{V_1} \subseteq H_1$. This completes the proof of case (A). $\square$

Proof of Case (B). In this case, we have that $T = \mathbb{Z}$ and $n = 2$.

Let $T = \langle t \rangle$ for some generator t , and let $U_m = \{t^i \mid i \leq m\}$. Since for each $k \in \mathbb{Z}$ and each of E_0, E_1 , one of $E_it^k \subseteq E_i$, $E_it^k \subseteq E_i^*$, $E_i * t^k \subseteq E_i$, $E_i * t^k \subseteq E_i^*$ holds, we must in fact have that $E_i = U_{n_i}$ or $E_i^* = U_{n_i}$ for $i = 0, 1$ and some n_0, n_1 .

We can thus easily see that there is only one G -orbit of $(\sim)$ -equivalence classes of edges in ϵ_T , and indeed only one G -orbit of edges, and thus that the Bass–Serre construction in case (A) has only one vertex and one edge.

Now first suppose that $G = H * L$, with both H and L nontrivial. Consider the Bass–Serre tree X_G of this splitting. Restricting the almost invariant set E to $T = \langle t \rangle$ means exactly to look at the orbit of some base vertex $v \in V(X_G)$ under powers of t ,

$$\dots, t^{-1}v, v, tv, t^2v, \dots$$

and ask how this orbit crosses E . Since multiplication by t has no fixed points in this case, it has an axis in X_G .

The translation length of t is exactly the number of edges in the segment from v to tv . Each of these corresponds to a nontrivial component in the restriction of E . Thus the weight of T is the number of edge crossings in one period of the T -axis, i.e. the translation length.

In this case, the weight is 2, so t moves a vertex of the Bass–Serre tree across exactly two edges. Recalling the Bass–Serre tree of a free product, we see that t moves the vertex v from an H -vertex to an L -vertex and back to an H -vertex. Replacing t by t^{-1} if necessary, we get $T = \langle hl \rangle$ with $h \in H \setminus \{1\}$ and $l \in L \setminus \{1\}$. Thus $T = C(g_1g_2)$ with $g_1 \in H$ and $g_2 \in L$ as needed.

A similar argument applies in the HNN case: here, there is only one vertex orbit, but t still has a translation length of 2, and we see that $T = C(pg_1p^{-1}g_2)$ for some $g_1, g_2 \in H$. $\square$

9 Proof of the Main Theorem

Definition 9.1. Let G be a finitely generated group. Then the **first Betti number** of G is defined as $\beta_1(G) = \text{rk } H_1(G; \mathbb{Z}) = \text{rk}(G_{ab})$, the rank of the abelianization $G_{ab} = G/[G, G]$ as a finitely generated abelian group.

The following basic property of β_1 is what will be of use:

Proposition 9.1. *A finitely generated group G has $\beta_1(G) > 0$ if and only if there exists a surjective homomorphism $G \rightarrow \mathbb{Z}$.*

Proof. If $\beta_1(G) > 0$, then G_{ab} is of positive rank, so projection onto an infinite cyclic factor gives a surjection $G_{ab} \rightarrow \mathbb{Z}$. Then the composition $G \rightarrow G_{ab} \rightarrow \mathbb{Z}$ is a surjection. Conversely, given such a surjection $G \rightarrow \mathbb{Z}$, passing to the abelianization of both G and $\mathbb{Z}$ gives a surjection $G_{ab} \rightarrow \mathbb{Z}_{ab} = \mathbb{Z}$, and thus shows that $\beta_1(G) > 0$. $\square$

The following theorem was proved in [9] using methods rather dissimilar to those used in the rest of the proof of the main theorem. As such, a detailed account of the proof is not given here.

Theorem 9.1 (Theorem 1 in [9]). *If G is a PD^2 group, then $\beta_1(G) > 0$.*

Lemma 9.1. *Let G be a group of type FP_2 . Then there exists a short exact sequence of groups*

$$0 \rightarrow M \rightarrow X \rightarrow G \rightarrow 0$$

such that X is finitely presented and $M_{ab} = 0$.

Proof. Since G is of type FP_1 , it is finitely generated. Then we may write $G = F_S/N$, where F_S denotes the free group on a finite set S and N is a normal subgroup of F_S . Since G is of type FP_2 , N_{ab} is a finitely generated $\mathbb{Z}G$ -module (see [4], Exercise VIII.5.3), so we can choose generators $\{r_1, \dots, r_m\}$ such that $\bar{r}_1, \dots, \bar{r}_m$ generate N_{ab} as a $\mathbb{Z}G$ -module. Let $A = \langle\langle r_1, \dots, r_m \rangle\rangle_{F_S}$ and let $X = F_S/A$. Since $A \subseteq N$, we have a natural surjection $X \rightarrow G$ with kernel N/A . X is clearly finitely presented, and by construction,

$$(N/A)_{ab} = (N/A)/[N/A, N/A] \cong N/(A[N, N]) = 0,$$

so we are done. $\square$

Lemma 9.2. *Let G be a group of type FP_2 admitting an infinite cyclic factor group. Then G splits nontrivially over a finitely generated subgroup.*

Proof. This proof is adapted from [3], which proves this fact in a more general setting. Following Lemma 9.1, let $1 \rightarrow M \rightarrow X \xrightarrow{\pi} G \rightarrow 1$ be a short exact sequence such that X is finitely presented and M^{ab} is trivial. Let $t \in G$ and $H \triangleleft G$ be such that G/H is infinite cyclic generated by tH . Let $\bar{x}$ be such that $\pi(\bar{x}) = t$, and let $\bar{a}_1, \dots, \bar{a}_m$ generate X . By replacing $\bar{a}_i$ with $\bar{a}_i \bar{x}^{-k_i}$ if necessary (where $\pi(\bar{a}_i)N = t^{k_i}N$), this set can be chosen such that $\pi(\bar{a}_i) \in H$ for all i . Thus X is presented as

$$X = F/R = \langle a_1, \dots, a_m, x \mid r_1, \dots, r_n \rangle$$

for some finite set of r_i , where $\bar{a}_i = a_i R$, $\bar{x} = xR$.

Since $\pi(r_i R)N = N$ for all i , and each $\pi(\bar{a}_i)$ is in N , each r_i must have exponent sum zero in x . Indeed, then, there must exist a natural number λ such that all $r_1, \dots, r_n$ are contained in the finitely generated subgroup

$$U = \langle x^k a_i x^{-k} \mid 1 \leq i \leq m, -\lambda \leq k \leq \lambda \rangle \subseteq F.$$

Consider the subgroup

$$K = \langle t^k \pi(\bar{a}_i) t^{-k} \mid 1 \leq i \leq m, -\lambda \leq k < \lambda \rangle \subseteq G.$$

Define $B = \pi(UR/R)$, let $G' = \langle B \rangle$ (so that $K \subseteq G'$), and let G^* denote the HNN extension

$$G^* = G' *_{K \sim K^t, p} = \langle B, p; psp^{-1} = tst^{-1} \rangle$$

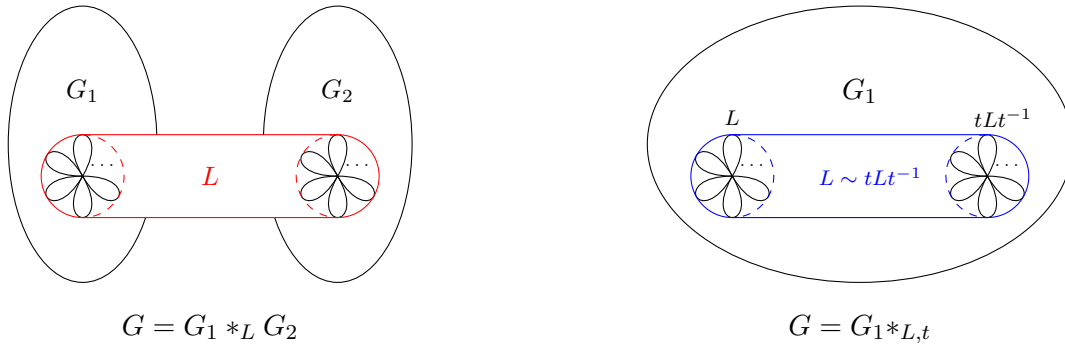
Since $\pi(a_1), \dots, \pi(a_m) \in B$, we have an epimorphism $\psi : G^* \rightarrow G$ induced by the inclusion $B \subseteq G$ and $\psi(p) = t$. We can also define an epimorphism $\varphi : F \rightarrow G^*$ by $\varphi(a_i) = \pi(\bar{a}_i)$ and $\varphi(x) = p$, and since φ sends each relator r_i to the identity in G^* , we have an induced map $\bar{\varphi} : X = F/R \rightarrow G^*$. The map $\psi \circ \bar{\varphi}$ coincides with π , and so $\ker(\psi \circ \bar{\varphi}) = M$. Then we know that $(\ker(\psi \circ \bar{\varphi}))_{\text{ab}} = 1$, whence $(\ker \psi)_{\text{ab}} = 1$ (since $\bar{\varphi}$ is surjective). However, $\ker \psi$ is a normal subgroup of the HNN extension G^* , and $\ker \psi \cap B = 1$, so indeed $\ker \psi$ must be a free group by Theorem 6 of [12]. Since nontrivial free groups are not perfect, $\ker \psi = 1$, so ψ is an isomorphism and $G \cong G^*$. $\square$

Corollary 9.1. *Let G be a PD^2 group. Then G splits nontrivially over a finitely generated free subgroup.*

Proof. By Theorem 9.1, $\beta_1(G) > 0$. Thus Proposition 9.1 tells us that there exists a surjection $G \rightarrow \mathbb{Z}$, giving an infinite cyclic factor group of G . Since PD^2 groups are of type FP (and thus certainly of type FP_2 , see Proposition 4.2), we may apply Lemma 9.2.

We thus know that G splits nontrivially as $G = G_1 *_L G_2$ or $G_1 *_L, t$ over some finitely generated subgroup L . The group L is of infinite index in G as an edge group in a nontrivial splitting of G , so by Corollary 4.2, L is free. $\square$

By taking the realization of the graph of spaces associated with this splitting, we get a $K(G, 1)$ of the following form:



The group G splits over a finitely generated free group L .

In the above figure and the figures that follow, a red band will denote an amalgamated product, and a blue band will denote an HNN extension.

Lemma 9.3. *Every PD^2 group splits nontrivially over an infinite cyclic subgroup.*

Proof. Let G be a PD^2 group. By Corollary 9.1, G splits nontrivially over a free subgroup L of finite rank. If $\text{rk } L = 1$, then we are done. Suppose that $\text{rk } L \geq 2$. Then either $G = G_1 *_L G_2$ for some subgroups G_1, G_2 or $G = G_1 *_L, t$ for some subgroup G_1 and some stable letter t .

First assume that $G = G_1 *_L G_2$. Consider the Mayer-Vietoris sequence

$$\cdots \rightarrow 0 \rightarrow H^1(G; \mathbb{Z}G) \oplus H^1(G_2; \mathbb{Z}G) \xrightarrow{(\text{res}_1, -\text{res}_2)} H^1(L; \mathbb{Z}G) \xrightarrow{\delta} H^2(G; \mathbb{Z}G) \rightarrow \cdots \quad (9.1)$$

Since $H^2(G; \mathbb{Z}G) = \mathbb{Z}$ and L has infinitely many ends (as a free group of rank > 1), the restriction of δ to $H^1(L; \mathbb{Z}L)$ cannot be injective, which by exactness means that $H^1(L; \mathbb{Z}L) \cap \text{im}(\text{res}_1, -\text{res}_2) \neq 0$. Since res_1 and res_2 are injective, the weight of L with respect to both $(G_1; \emptyset)$ and $(G_2; \emptyset)$ cannot be zero.

Now assume for contradiction that the weight of L with respect to both $(G_1, \emptyset)$ and $(G_2, \emptyset)$ is greater than one. Let $C_L = H^1(L; \mathbb{Z}L)$ and $C_i = H^1(G_i; \mathbb{Z}G_i)$ for $i = 1, 2$. Choose coset representatives $\{x_i : i \in I\}$ and $\{y_j : j \in J\}$ for the right cosets of G_1/L and G_2/L respectively, with both sets containing the identity. Then, by the normal form for amalgamated free products, the coset representatives for G/G_1 are reduced alternating words in G_1/L and G_2/L ending in a G_2 -letter.

Thus L has weight 1 with respect to $(G_1; \emptyset)$ or $(G_2; \emptyset)$, say $(G_1, \emptyset)$. Then Theorem 8.1 (A) gives that either

1. $G_1 = H_1 * H_2$, $L = L_1 * L_2$, $e \neq L_i \subset H_i$ for $i = 1, 2$.
2. $G_1 = H_1 * \langle p \rangle$, $L = L_1 * pL_2p^{-1}$, $e \neq L_1, L_2 \subset H$.

In the first case,

$$\begin{aligned} G &= (H_1 * H_2) *_L G_2 \\ &= \langle H_1, H_2, G_2 \mid i_{H_1}(x) = i_{G_2}(x), i_{H_2}(y) = i_{G_2}(y), x \in L_1, y \in L_2 \rangle \\ &= H_1 *_L (H_2 *_L G_2). \end{aligned}$$

If $H_1 \neq L_1$, then this gives a nontrivial splitting of G over L_1 . If $H_1 = L_1$, then $H_2 \neq L_2$, and the expression above gives a nontrivial splitting of G over L_2 .

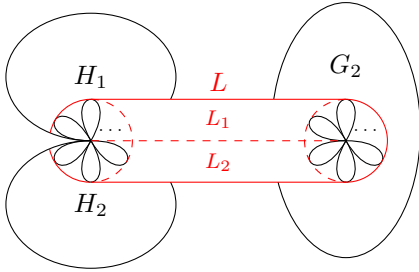


Figure 3: $G = (H_1 * H_2) *_L G_2$

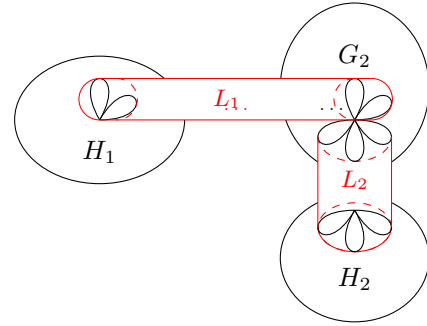


Figure 4: $G = H_1 *_L (H_2 *_L G_2)$

A geometric picture showing that the two splittings are equivalent.

If $H_1 = L_1$, then $H_2 \neq L_2$ and $G = H_2 *_L G_2$ splits over L_2 . In the second case, we have that

$$\begin{aligned} G &= (H_1 * \langle p \rangle) *_L G_2 \\ &= \langle (H_1 * \langle p \rangle), G_2 \mid i_{G_1}(x) = i_{G_2}(x); x \in L_1 * pL_2p^{-1} \rangle \\ &= \langle H_1, G_2, p \mid i_{H_1}(x) = i_{G_2}(x), p^{-1}i_{H_1 * \langle p \rangle}(y)p = p^{-1}i_{G_2}(y)p; x \in L_1, y \in L_2 \rangle \\ &= \langle H_1 *_L G_2, p \mid p^{-1}i_{(H_1 *_L G_2) * \langle p \rangle}(y)p = p^{-1}i_{(H_1 *_L G_2) * \langle p \rangle}(y)p; y \in L_2 \rangle \\ &= (H_1 *_L G_2) *_L p, \end{aligned}$$

and this splitting is clearly nontrivial. Therefore G splits over either L_1 or L_2 in both cases, so in particular G splits over a free group of strictly lower rank than L .

The case where $G = G_1 *_L t$ has a similar proof. Using the normal form for an HNN extension, one again sees that L has weight 1 with respect to the relevant pair, and Theorem 8.1(A) produces a splitting

$L = L_1 * L_2$. Then Theorem 8.1 (A) gives the same cases (1) and (2), as above, and we can change the stable letter t such that $tLt^{-1} \subseteq H_1$. In the first case, we have that

$$\begin{aligned}
G &= (H_1 * H_2) *_{L,t} \\
&= \langle (H_1 * H_2), t \mid txt^{-1} = x; x \in L \rangle \\
&= \langle H_1, H_2, t \mid txt^{-1} = x, tyt^{-1} = y; x \in L_1, y \in L_2 \rangle \\
&= \langle (H_1 *_{L_1,t}), H_2 \mid t^{-1}yt = y; y \in L_2 \rangle \\
&= (H_1 *_{L_1,t}) *_{L_2} H_2.
\end{aligned}$$

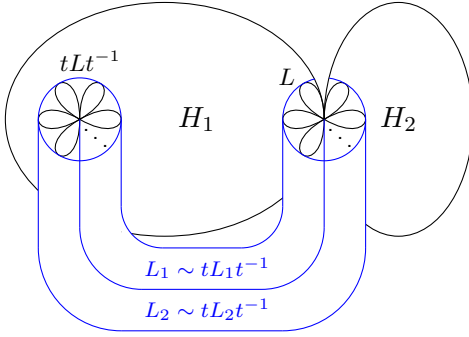


Figure 5: $G = (H_1 * H_2) *_{L,t}$

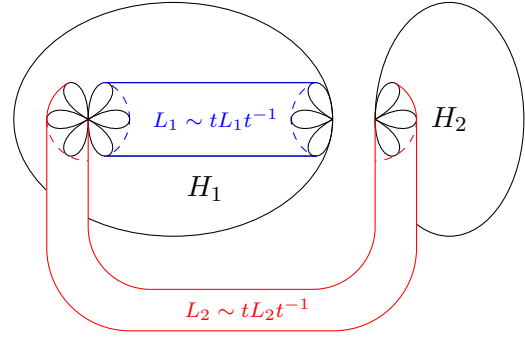


Figure 6: $G = (H_1 *_{L_1,t}) *_{L_2} H_2$

A geometric picture showing that the two splittings are equivalent.

This splits over L_2 if $L_2 \neq H_2$, and otherwise splits over L_1 by the same steps, as in the previous case. Similarly, in the second case, we have $G = (H_1 *_{1,p}) *_{L,t} = (H_1 *_{L_1,p}) *_{L_2,t^{-1}}$.

In any case, then, G splits nontrivially over a free group L_1 or L_2 , where $L = L_1 * L_2$ is a nontrivial splitting of L . Since the splitting is nontrivial, both L_1 and L_2 are of strictly smaller rank than L . Repeating this process, the rank of the free edge group strictly decreases, and so this process terminates with a splitting over a free subgroup of rank 1. $\square$

We will now use the theory of duality pairs to realize this splitting as an operation producing a closed surface.

Definition 9.2. Let $(G; \mathcal{S})$ be a group pair. Then $(G, \mathcal{S})$ is called **geometric** if it is of the following form:

- (a) G is the fundamental group of a closed surface with $m + 1$ discs removed (where $m \geq 0$, or $m \geq 1$ if the surface is a sphere).
- (b) $\mathcal{S} = \{S_0, \dots, S_m\}$, where $S_0, S_1, \dots, S_m$ are the infinite cyclic subgroups of G represented by its boundary components.

Following Equation (2.1), we can easily see that a group pair $(G; S_0, \dots, S_m)$ is geometric if it is of one of the following forms:

1. G is the free group on the elements $t_1, \dots, t_m, x_1, y_1, \dots, x_g, y_g$ (for $m + g > 0$), $S_1, \dots, S_m$ are generated by elements conjugate to $t_1, \dots, t_m$, and S_0 is generated by $t_1, \dots, t_m \cdot \prod_{i=1}^g [x_i, y_i]$. This corresponds to the case of the orientable surface of genus g . Call this form $\mathcal{O}(m, g)$.

2. G is the free group on the elements $t_1, \dots, t_m, z_0, \dots, z_k$ (for $m, k \geq 0$), $S_1, \dots, S_m$ are generated by elements conjugate to $t_1, \dots, t_m$, and S_0 is generated by $t_1, \dots, t_m \cdot \prod_{i=1}^k z_i^2$. This corresponds to the case of the non-orientable surface of non-orientable genus k . Call this form $\mathcal{N}(m, g)$.

Proposition 9.2. *Let G be a PD^2 group that splits nontrivially over a cyclic group C as either $G = G_1 *_C G_2$ or $G = G_1 *_C p$. Then in the first case, the pairs $(G_1; C)$ and $(G_2; C)$ are both PD^2 pairs. In the second case, the pair $(G_1; C, p^{-1}Cp)$ is a PD^2 pair.*

Proof. The first case follows from Proposition 5.4, the fact that G is a PD^2 group, and the fact that C is a PD^1 group. The second case follows from Proposition 5.5 for the same reasons. $\square$

The main theorem thus follows from the following statement:

Theorem 9.2. *Every PD^2 pair is geometric.*

Proof. Let $(G; S_0, S_1, \dots, S_m)$ be a PD^2 pair. By Theorem 5.2, we know that G is finitely generated free and that the S_i are cyclic groups. Then Proposition 5.3 (4) says that the fundamental class $e \in H_2(G, \mathcal{S}; \tilde{\mathbb{Z}})$ determines the fundamental classes e_i of the cyclic groups $S_0, \dots, S_m$, and in particular that the e_i are the components of $\partial e \in H_1(\mathcal{S}; \tilde{\mathbb{Z}}) = \oplus_{i=0}^m H_1(S_i; \tilde{\mathbb{Z}})$, where ∂ is the connecting homomorphism in the long exact sequence in homology of G modulo $\mathcal{S}$. By the cap product ladder of Theorem 5.1, we see that the diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & H^1(G; \mathbb{Z}G) & \xrightarrow{\{\text{res}_i\}} & \oplus_{i=0}^m H^1(S_i; \mathbb{Z}G) & \xrightarrow{\delta} & H^2(G, \mathcal{S}; \mathbb{Z}G) \longrightarrow 0 \\
& & & & \downarrow \{e_i \cap -\} & & \downarrow (e \cap -) \\
& & & & \oplus_{i=0}^m H_0(S_i; \tilde{\mathbb{Z}} \otimes \mathbb{Z}G) & \xrightarrow{\text{cor}} & H_0(G; \tilde{\mathbb{Z}} \otimes \mathbb{Z}G) \\
& & & & \downarrow j & & \downarrow \cong \\
& & & & \oplus_{i=0}^m (\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G) & \xrightarrow{p} & \tilde{\mathbb{Z}}
\end{array}$$

commutes, where the top row is exact, the cap product maps are isomorphisms by duality, and $p(1 \otimes_{S_i} y) = 1 \cdot y$ for $y \in G$.

We now induct on the rank of G . If G is infinite cyclic, then $H^1(G; \mathbb{Z}G)$ is infinite cyclic, as is $H^2(G, \mathcal{S}; \mathbb{Z}G)$, so in fact

$$\oplus_{i=0}^m H^1(S_i; \mathbb{Z}G) \cong \oplus_{i=0}^m (\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G)$$

is free abelian of rank 2. Note that in this case $G = \langle t \rangle$ and $S_i = \langle t^{m_i} \rangle$ for all i . so $\mathbb{Z}G \cong \mathbb{Z}[t, t^{-1}]$ and G acts on $\tilde{\mathbb{Z}}$ either trivially or via $t^k n = (-1)^k n$. In the first case, $\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G \cong \mathbb{Z}[G/S_i] \cong \mathbb{Z}[\mathbb{Z}/m_i \mathbb{Z}] \cong \mathbb{Z}^{m_i}$, so to satisfy the criterion either $m = 1$ and $S_0 = S_1 = G$ or $m = 0$ and $S_0 = \langle t^2 \rangle$ has index 2 in G . The $m = 1$ case makes the pair an $\mathcal{O}(1, 0)$ pair, and the $m = 0$ case makes the pair an $\mathcal{N}(0, 1)$ pair.

In the case of a nontrivial G -action, we have that

$$\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G \cong \mathbb{Z}^{m_i}$$

if m_i is even and

$$\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G \cong (\mathbb{Z}/2\mathbb{Z})^{m_i}$$

if m_i is odd. Thus the only possibility is $S_0 = S_1 = G$ and $m = 0$ as before. This shows that $(G, \mathcal{S})$ is geometric if $\text{rk } G = 1$.

Now suppose that $\text{rk}(G) \geq 2$. Set $T = S_0$. We wish to determine the weight $n(T)$ of T with respect to the pair $(G; S_1, \dots, S_m)$. First, to show that $N(G; \mathcal{S}) \neq 0$, we see that this is true if and only if there exists an element of the form $(d, 0, 0, \dots, 0)$ in $\text{im}\{\text{res}_i\}$ for some $d \neq 0$. Since

$$\text{im}\{\text{res}_i\} = \ker \delta = \ker(p \circ j \circ \{e_i \cap -\})$$

by the diagram above, and j and $\{e_i \cap -\}$ are isomorphisms, this will happen only if the map

$$p : \tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G \xrightarrow{p} \tilde{\mathbb{Z}}$$

has nontrivial kernel. It clearly does: by the pigeonhole principle (since the index of T in G is more than 3, by the Schreier index formula), we can choose coset representatives g, g' of T in G such that $1 \cdot g = 1 \cdot g'$. Then $p(1 \otimes g - 1 \otimes g') = 0$, but $1 \otimes g \neq 1 \otimes g'$. Thus we get that $(G; S_1, \dots, S_m)$ is adapted.

Note that $H^1(T; \mathbb{Z}G) = \oplus_{x_v \in G/T} H^1(T; \mathbb{Z}T)x_v$. Now we consider the map

$$p \circ j \circ \{e_i \cap -\} : H^1(T; \mathbb{Z}T)x_v \rightarrow \tilde{\mathbb{Z}},$$

for each x_v in G/T . Then restricting $\tilde{\mathbb{Z}} \otimes_{S_i} \mathbb{Z}G$ only to the elements of $\mathbb{Z}G$ in the coset Tx_v recovers an isomorphic copy of $\tilde{\mathbb{Z}}$ (every element can be put in the form $k \otimes x_v$ for some k), so p is a bijection on this restricted set. Indeed, then, any $d \neq 0$ with nontrivial component in only one of the $H^1(T; \mathbb{Z}T)x_v$ will not be mapped to zero under p . However, the process above showed how to find a d with nontrivial components in exactly two of the $H^1(T; \mathbb{Z}T)x_v$, so we conclude that the weight of T in $(G; S_1, \dots, S_m)$ is exactly two.

By Theorem 8.1 (B), we are in one of the following cases:

1. $G = G_1 * G_2, T = S_0 = C(g_1 g_2)$, $e \neq g_i \in G_i$ for $i = 1, 2$, and the subgroups $S_1, \dots, S_k$ are conjugate to subgroups of G_1 , while $S_{k+1}, \dots, S_m$ are conjugate to subgroups of G_2 for some $0 \leq k \leq m$.
2. $G = G_1 * \langle p \rangle$, $T = S_0 = C(p g_1 p^{-1} g_2)$, $e \neq g_1, g_2 \in G_1$ and $S_1, \dots, S_m$ are conjugate to subgroups of G_1 .

By replacing the S_i with conjugates if necessary, we may assume that they are each honest subgroups of G_1 or G_2 .

In case (1), we can write

$$G = (G_1 * G_2) = (G_1 * C(g_2)) *_{C(g_2)} G_2,$$

such that $S_0, S_1, \dots, S_k$ are in $G_1 * C(g_2)$, and $S_{k+1}, \dots, S_m$ are in G_2 . Then Proposition 5.4 gives that $(G_2; C(g_2), S_{k+1}, \dots, S_m)$ is a PD^2 pair, and $(G_1; C(g_1), S_1, \dots, S_k)$ is as well.

In case (2), we can write

$$\begin{aligned} G &= G_1 * \langle p \rangle = \langle G_1, p \mid \emptyset \rangle \\ &= \langle G_1, p, a \mid a = p g_1 p^{-1} g_2 \rangle \\ &= \langle G_1 * \langle a \rangle, p \mid p^{-1} a g_2^{-1} p = g_1 \rangle \\ &= (G_1 * \langle a \rangle) *_{C(a g_2^{-1}) \sim C(g_1), p}, \end{aligned}$$

so that $S_0 = C(a)$, $S_1, \dots, S_m$ are in $G_1 * C(a)$. Then Proposition 5.5 gives that $(G_1; S_1, \dots, S_m, C(g_1), C(g_2))$ is a PD^2 pair.

Thus, given a PD^2 pair $(G, \mathcal{S})$, we can express G as an amalgamation or HNN extension over one or two groups of strictly lower rank with associated PD^2 pairs, where the amalgamation or extension is taken over $S_0 \in \mathcal{S}$. By the inductive hypothesis, we can assume that these pairs are geometric. It is clear geometrically that amalgamating or taking an HNN extension along the removed holes in a closed surface with holes removed gives another closed surface with holes removed, and thus that the amalgamation along S_0 is a closed surface. Therefore, by induction, we see that the original pair $(G, \mathcal{S})$ is geometric. $\square$

Theorem 9.3. *Every PD^2 group is a surface group.*

Proof. Let G be a PD^2 group. By Lemma 9.3, G admits a nontrivial splitting over an infinite cyclic subgroup C . Thus G is either an amalgamated product

$$G = G_1 *_C G_2$$

or an HNN extension

$$G = G_1 *_C p.$$

First suppose that

$$G = G_1 *_C G_2.$$

Proposition 9.2 implies that $(G_1; C)$ and $(G_2; C)$ are both PD^2 pairs. By Theorem 9.2, both of these pairs are geometric. Hence there exist compact connected surfaces M_1 and M_2 , each obtained from a closed aspherical surface by removing one open disc, such that

$$\pi_1(M_i) \cong G_i$$

and the subgroup $C \leq G_i$ is represented by the boundary component ∂M_i . The amalgamated product $G_1 *_C G_2$ is precisely the group obtained by identifying these two boundary subgroups. Geometrically, this corresponds to gluing the boundary circle of M_1 to the boundary circle of M_2 . Let M be the closed surface obtained by this gluing. By the Seifert–van Kampen theorem,

$$\pi_1(M) \cong G_1 *_C G_2 \cong G.$$

Since G is a PD^2 group, it is torsion-free. Thus M is not S^2 or $\mathbb{RP}^2$, and so M is aspherical. Hence G is a surface group.

Now suppose that

$$G = G_1 *_C p.$$

Again C is infinite cyclic and hence a PD^1 group. By Proposition 9.2, the pair

$$(G_1; C, p^{-1}Cp)$$

is a PD^2 pair. By Theorem 9.2, this pair is geometric. Hence there is a compact connected surface M_1 obtained from a closed aspherical surface by removing two open discs, such that

$$\pi_1(M_1) \cong G_1$$

and the two boundary components represent the subgroups C and $p^{-1}Cp$.

The HNN extension identifies these two boundary subgroups. Geometrically, this corresponds to gluing the two boundary circles of M_1 to each other. Let M be the closed surface obtained by this gluing. By the Seifert–van Kampen theorem,

$$\pi_1(M) \cong G_1 *_C p \cong G.$$

As above, G is torsion-free, so M is not S^2 or $\mathbb{RP}^2$. Therefore M is aspherical, and hence G is a surface group. This proves that every PD^2 group is a surface group. $\square$

10 Applications / Further Questions

The main theorem has an immediate corollary concerning so-called *Poincaré 2-complexes*.

Definition 10.1. Let X be a connected CW complex with $G = \pi_1(X)$ that is dominated by a finite CW complex P (in the sense that there exist maps $\lambda : X \rightarrow P$, $\lambda' : P \rightarrow X$ such that $\lambda'\lambda \cong \text{id}_X$). Suppose we are given a homomorphism $w : G \rightarrow \{\pm 1\}$ defining a G -module structure $\tilde{\mathbb{Z}}$ on $\mathbb{Z}$, an integer n , and a class $[X] \in H_n(X; \tilde{\mathbb{Z}})$ such that the cap product with $[X]$ induces an isomorphism

$$[X] \frown (-) : H^r(X; A) \rightarrow H_{n-r}(X; A \otimes \tilde{\mathbb{Z}})$$

for all G -modules A . In this case, we called X a (connected) **Poincaré duality complex** of dimension n , or a *Poincaré n -complex*.

Corollary 10.1. An arbitrary CW complex is a Poincaré 2-complex if and only if it is homotopy equivalent to a closed surface.

Proof. In [19], Wall showed that a Poincaré 2-complex X whose fundamental group $G = \pi_1(X)$ is finite is homotopy equivalent to S^2 or to $\mathbb{RP}^2$; it was additionally shown that if $\pi_1(X)$ is infinite, then X is aspherical.

This exactly means that X is a $K(G, 1)$ complex; the (co)homology of G and X therefore coincide, and so the duality condition implies that G is a PD^2 group. Therefore there exists a closed surface Y that is also a $K(G, 1)$ complex. Since all $K(G, 1)$ complexes are homotopy equivalent, this completes the proof. $\square$

The question of course takes on a very different flavor in dimensions higher than 2. In particular, it was proven in [7] that in dimensions $n \geq 4$ there exist PD^n groups that are not finitely presented. As every compact manifold has finitely presented fundamental group, such groups cannot be the fundamental groups of closed manifolds. The question is entirely open in dimension 3.

References

- [1] R. Bieri and B. Eckmann. Groups with homological duality generalizing Poincaré duality. *Inventiones Mathematicae*, 20:103–124, 1973.
- [2] R. Bieri and B. Eckmann. Relative homology and Poincaré duality for group pairs. *Journal of Pure and Applied Algebra*, 13:257–319, 1978.
- [3] R. Bieri and R. Strebel. Almost finitely presented soluble groups. *Commentarii Mathematici Helvetici*, 53:258–278, 1978.
- [4] K. S. Brown. *Cohomology of Groups*, volume 87 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1982.
- [5] Kenneth S. Brown. Homological criteria for finiteness. *Commentarii Mathematici Helvetici*, 50:129–135, 1975.
- [6] Daniel E. Cohen. *Groups of Cohomological Dimension One*, volume 245 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, Heidelberg, New York, 1972.
- [7] M. W. Davis. The cohomology of a Coxeter group with group ring coefficients. *Duke Mathematical Journal*, 91(2):297–314, 1998.

- [8] M. J. Dunwoody. Accessibility and groups of cohomological dimension one. *Proceedings of the London Mathematical Society*, 38:193–215, 1979.
- [9] B. Eckmann and P. Linnell. Poincaré duality groups of dimension two, ii. *Commentarii Mathematici Helvetici*, 58:111–114, 1983.
- [10] B. Eckmann and H. Müller. Poincaré duality groups of dimension two. *Commentarii Mathematici Helvetici*, 1980.
- [11] Allen Hatcher. *Algebraic Topology*. Cambridge University Press, Cambridge, 2002.
- [12] A. Karrass and D. Solitar. Subgroups of hnn groups and groups with one defining relation. *Canadian Journal of Mathematics*, 23(4):627–643, 1971.
- [13] H. Müller. Decomposition theorems for group pairs. *Mathematische Zeitschrift*, 176:223–246, 1981.
- [14] Jean-Pierre Serre. *Trees*. Springer-Verlag, Berlin, Heidelberg, New York, 1980.
- [15] J. Stallings. *Group Theory and Three-Dimensional Manifolds*. Yale University Press, New Haven and London, 1971.
- [16] J. R. Stallings. On torsion-free groups with infinitely many ends. *Annals of Mathematics*, 88(2):312–334, 1968.
- [17] R. Strebel. A remark on subgroups of infinite index in poincaré duality groups. *Commentarii Mathematici Helvetici*, 52:317–324, 1977.
- [18] R. G. Swan. Groups of cohomological dimension one. *Journal of Algebra*, 12:585–601, 1969.
- [19] C. T. C. Wall. Poincaré complexes: I. *Annals of Mathematics*, 86(2):213–245, 1967.
- [20] C. T. C. Wall, editor. *Homological Group Theory*. Cambridge University Press, 1979. p. 391.